

Ganglion :

m/c swelling at the wrist : **Dorsal** > Ventral
unilocular cystic swelling.

Origin : Synovium.

Contents : mucinous fluid.

Treatment :

- Conservative.
- Aspirate contents → Inject steroids/sclerosants.
- Surgical excision.

Ganglion cyst



----- Active space -----

Compound palmar ganglion :

misnomer.

Pathophysiology :

Chronic inflammation of flexor sheath of forearm & hand

↓
Swelling of flexor sheath except at the wrist
(D/t flexor retinaculum)

↓
Hour glass appearance of forearm & hand

Etiology → TB
→ Rheumatoid arthritis



Compound palmar ganglion

Conditions Around Knee

01:00:18

Osteochondritis dissecans :

Bone fragment breaks away at bone & cartilage interface

↓
Prevents gliding movement of knee joint (Locking of knee joint)

m/c site : **knee** > elbow

↓
Lateral part of medial femoral condyle

C/F → knee pain
→ Sudden locking of knee



CT



X ray

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Evaluation :

Wilson's test :

Pain on extension of internally rotated leg

↓
Relieved on external rotation of leg

Wilson's test

Loose bodies :

- m/c cause of loose body in knee joint : Osteophytes in osteoarthritis.
- m/c cause of loose body in knee —
 - In young : Osteochondritis dissecans.
 - In elderly : Osteophytes in osteoarthritis.
- m/c cause of multiple loose bodies in knee : Synovial chondromatosis

↓
metaplasia of the synovium↓
multiple cartilagenous bodies in joint↓
On Xray : Snowstorm appearance of knee

Synovial chondromatosis

Chondromalacia patellae :

- Softening of cartilage.
- m/c cause : Idiopathic.
- m/c cause of anterior knee pain.
- F > m.
- Aggravating factors : Getting up from sitting position after prolonged sitting (movie sign).



Normal

Chondromalacia patella

Baker's cyst :

Pulsion/pressure diverticulum of synovium.

Location : Posterior aspect of knee.

Etiology : Osteoarthritis, rheumatoid arthritis.

On examination : Prominent on knee extension,
disappears on knee flexion.

Diagnosis : USG.

Treatment : Excision.



Baker's cyst

Note :

Function of synovial sac :

- minimizes friction.
- Allows smooth gliding of tendons.

Bursitis :

Inflammation of synovial sac d/t repetitive loading.

Types :

1. Housemaid's knee



2. Clergyman's knee



3. Student's elbow



4. Haglund deformity



Chronic inflammation

Calcification : Bump formed

5. Tailor's bunion



6. Weaver's bottom : Ischial bursitis.

01:11:14

Conditions of Foot



Hallux valgus :

Lateral deviation of great toe → Swelling at first MTP → Causing pain

D/d for pain at first MTP : Gout.

- Pain d/t inflammation & destruction of first MTP joint.
- On knee aspiration : Needle-shaped monosodium urate crystals.

Treatment : Osteotomy.

Inflammation (+)
No valgus deformity



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Avascular Necrosis (AVN)

01:12:24

Bones with tendency of AVN :

- Head of femur (Spontaneous AVN in Chandler's disease).
- Proximal pole of scaphoid (Spontaneous AVN in Preiser's disease).
- Body of talus.

ETIOLOGY

Conditions associated with Idiopathic AVN :

- Sick cell disease.
- Gaucher's disease.
- Caisson's disease.
- Perthes' disease.
- Alcohol consumption.
- Corticosteroid use : Causes ↑ pressure in head of femur

↓ blood supply to head of femur → AVN → Pain at hip

CLINICAL FEATURES

- Young individuals : m > F.
 - B/L in {
 - 50% cases of AVN.
 - 80% in cases of AVN with h/o steroid use.
 - Pain at hip {
 - Limping : Antalgic gait
 - Radiates to knee
 - Restriction of movements : Internal rotation > Abduction
- ↓
Sectoral sign

INVESTIGATIONS

X-ray :

1. Incongruent/ deformed head with ↑ density (Takes weeks-months to appear).

a. Crescent sign :

- Early sign.
- D/t subchondral collapse of head of femur.

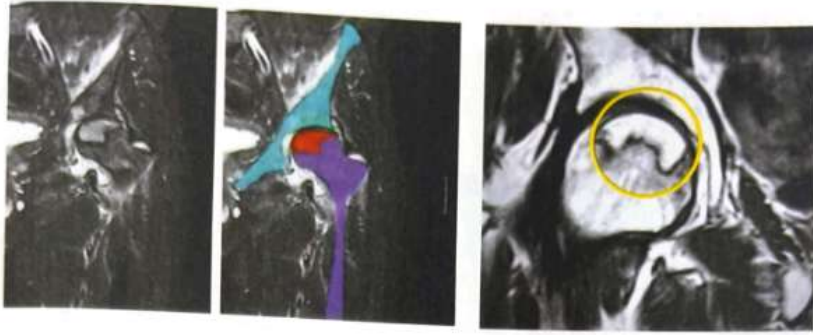


Incongruent head of femur



Crescent sign

MRI: IOC.



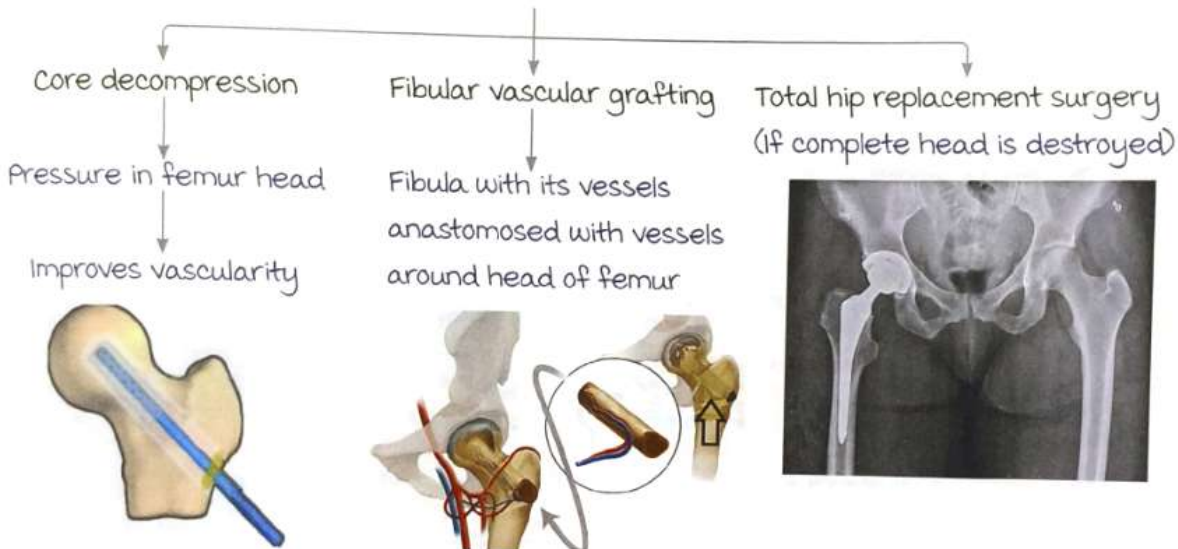
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Bone scan: Donut sign

- ↓ uptake of radionuclide by head of femur.
- Surrounding bone: Normal uptake.



TREATMENT



TRAUMATIC AVN

- Prolonged hip dislocation
 - Neck of femur fracture
- AVN of head of femur
- waist of scaphoid fracture
- AVN of proximal pole of scaphoid
- Neck of talus fracture
- AVN of body of talus



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ADDITIONAL RADIOLOGICAL FEATURES**Hawkin's sign :**

On X-ray : Subchondral lucency

↓
Indicates **viable bone** with adequate vascularity

Hawkin's sign

Snowcap sign :On X-ray of AVN of **head of humerus**.**Note :**

- Intertrochanteric fracture : ↑ risk of malunion.
- Hawkin's classification : For talus fractures.



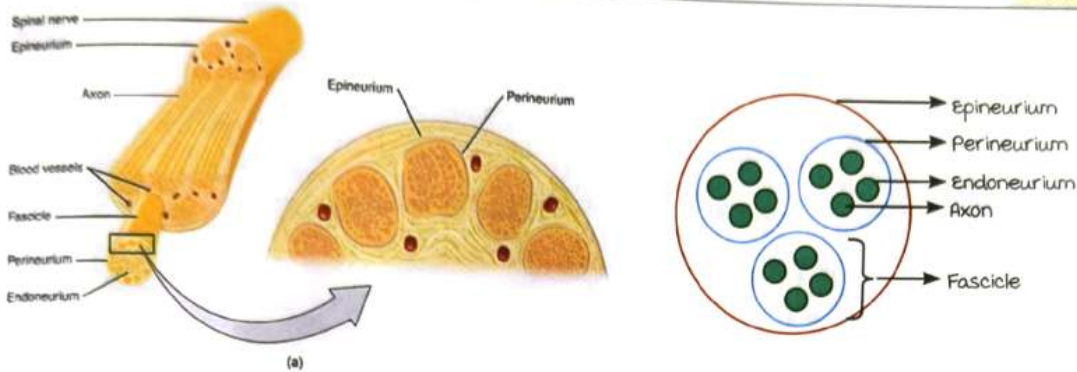
Snowcap sign

NERVE INJURIES : PART 1

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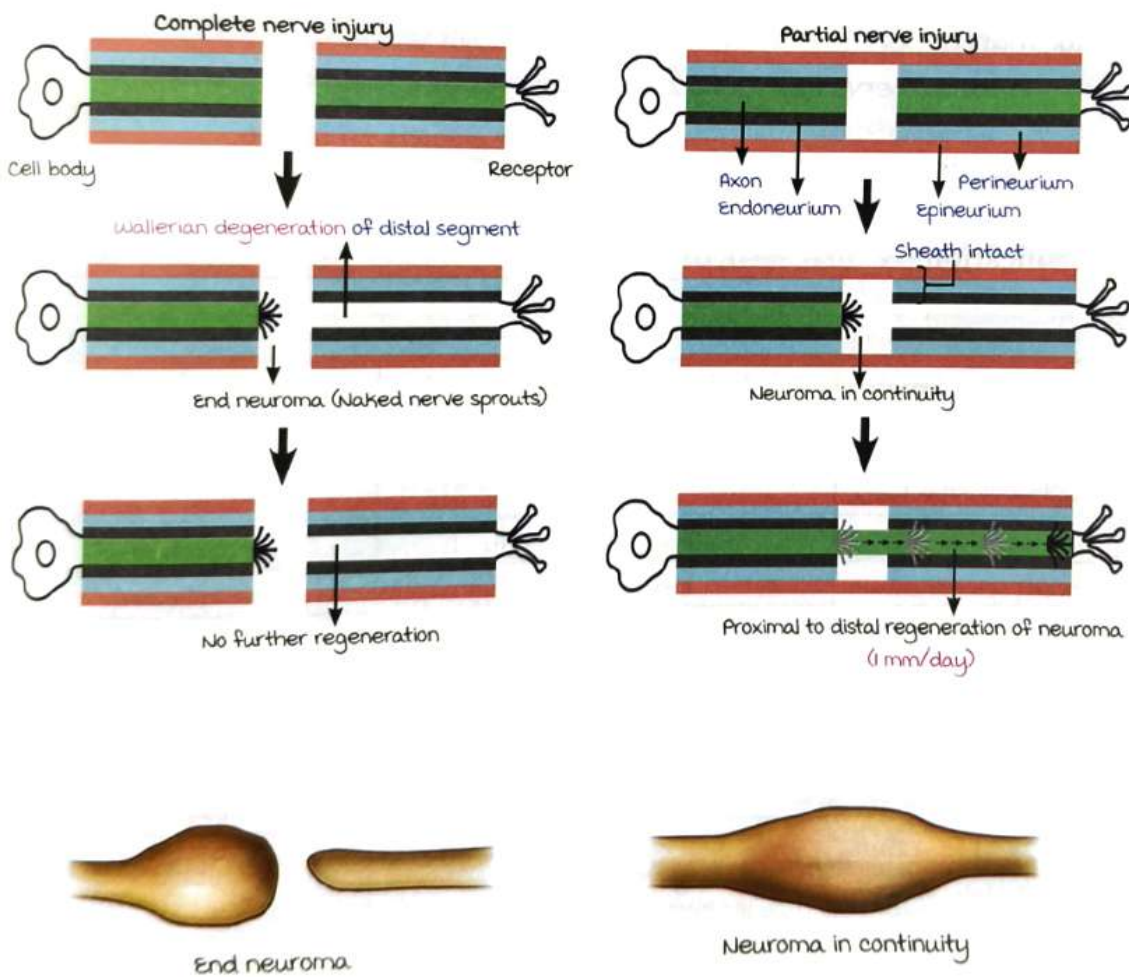
Structure of Nerve

00:01:56



Pathology in Nerve Injuries

00:02:58



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Tinel sign :

Distal to proximal percussion of nerve

Pain/ shock-like sensation in limb
d/t stimulation at neuroma (+)

Non-progressive

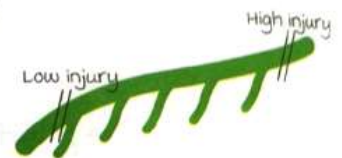
Progressive

End neuroma

Neuroma in continuity

High injury vs low injury :

- Higher (Proximal) the injury → more the disability/ deformity.
- Lower (Distal) the injury → Lesser the disability/ deformity.

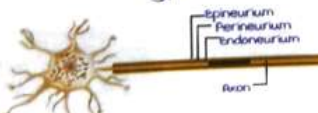
**motor march :**

In partial nerve injury : Recovery of nerve with neuroma in continuity

Gradual, progressive proximal to distal recovery of muscle functions

Classification

00:14:06

	Neuropraxia	Axonotmesis	Neurotmesis
Seddon's	Nerve compression ↓ Temporary reversible physiological conduction block	Axons injured with atleast one nerve sheath intact	Complete transection of nerve
	No neuroma	Neuroma in continuity	End neuroma
	100 % recovery spontaneously	< 100% recovery	No recovery
	-	Progressive +ve Tinel's	Non-progressive +ve Tinel's
	-	motor march (+)	
	Eg : Tourniquet palsy, saturday night palsy, crutch palsy.	Eg : Fracture, dislocations.	Eg : Lacerations, cut wound, incised wound.
Sunderland's	Type I 	Type II : Only axon injured Type III : Axon + endoneurium injured Type IV : Axon + endoneurium + perineurium injured	Type V 

Management

00:19:57

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DIAGNOSIS

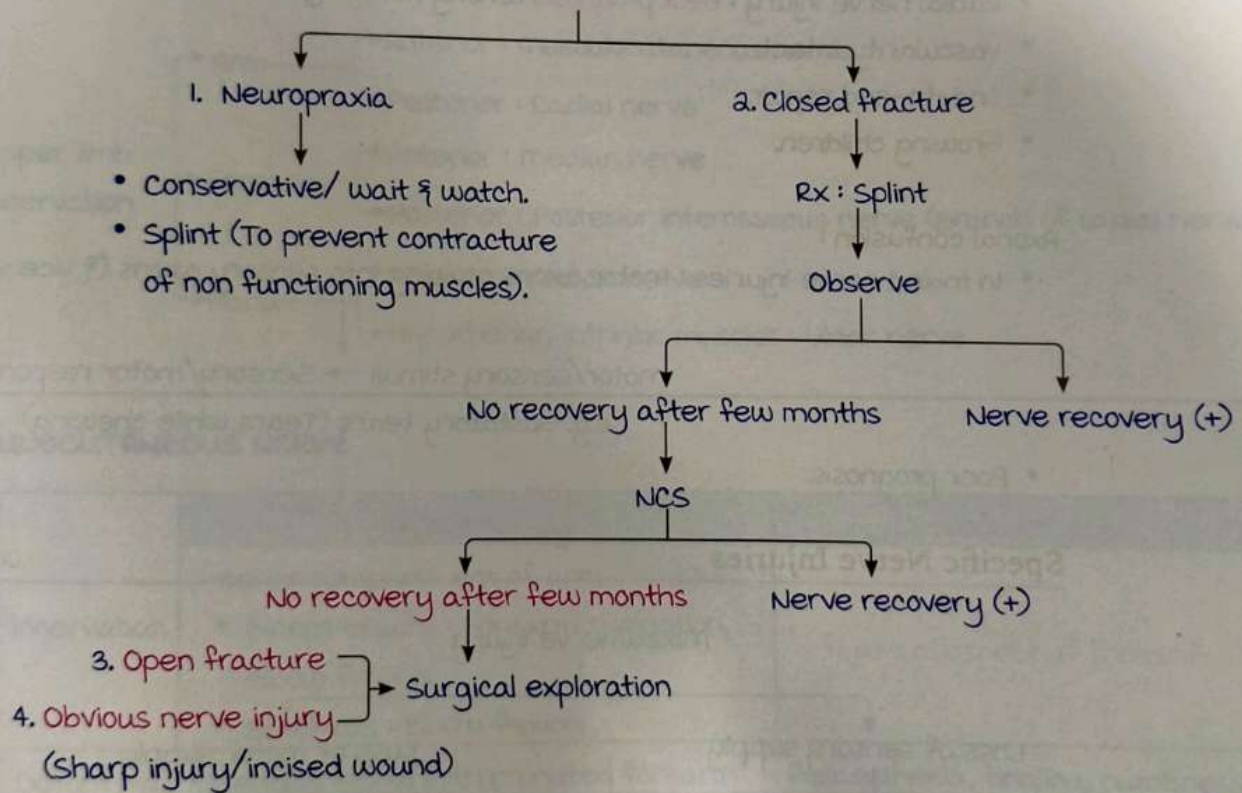
1. Tinel's sign (For **assessment of nerve recovery**).

2. Nerve conduction study :

- **IOC** to determine type of nerve injury.

- Parameters
 - Nerve conduction velocity (**Best**) : measures velocity of impulse transmitted through different segments of the nerve (Compression → ↓ velocity)
 - Electromyography

TREATMENT



NERVE REPAIR

- Types
- Primary : In < 6 hours
 - Delayed primary : In 7-18 days

- methods
- End-to-end repair
 - End-to-side repair

Nerve graft :

- Indication : If nerve ends do not approximate.
- Sources
 - Sural nerve (m/c)
 - Saphenous nerve

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Tendon transfer :

- Indication : Nerve injuries not repaired within **one year** of injury
 - ↓
 - motor end plate of the nerve die d/t lack of stimulation
 - ↓
 - Non-functional neuromuscular junction
- Procedure : Tendons & their muscle (Functional neuromuscular junctions (+)) repositioned to perform desired action.

Good prognostic factors :**NERVE - GROW**

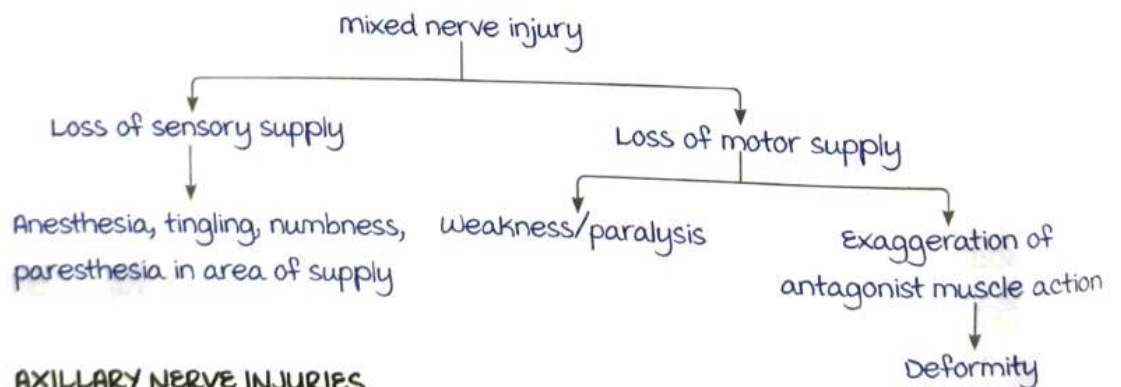
- Neuropraxia.
- Early repair.
- Radial nerve injury : **Best prognosis** among nerve injuries (motor nerve).
- Vascularity intact.
- End-to-end repair.
- Growing children.

Axonal confusion :

- In mixed nerve injuries : motor axons growing into sensory axons (& vice versa)
 - ↓
 - motor/sensory stimuli → Sensory/motor response
e.g. Gustatory tears (Tears while chewing)
- Poor prognosis.

Specific Nerve Injuries

00:26:45

**AXILLARY NERVE INJURIES**

Origin : Brachial plexus.
 Root value : C5, C6.
 mixed nerve.

	motor supply	Sensory supply
Innervation	• Deltoid (Abduction of shoulder). • Teres minor (External rotation of shoulder).	Lateral aspect of proximal shoulder
Deformity	Adducted & internally rotated	Regimental badge sign

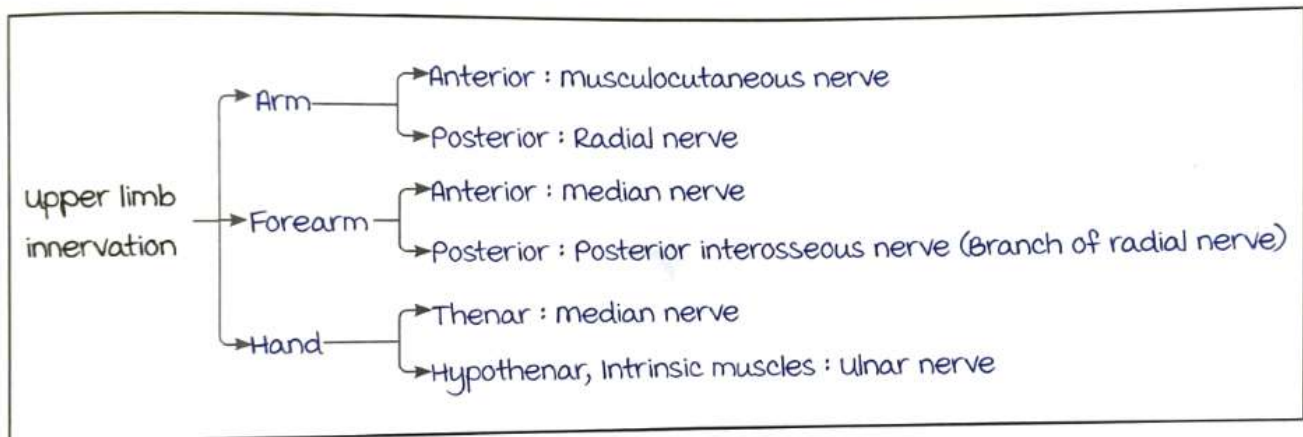
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mechanism of injury :

- **Shoulder dislocation** (m/c).
- Proximal humerus fracture
 - Surgical neck fracture
 - Glenoid fractures
- Iatrogenic.

Note :**Shoulder abduction :**

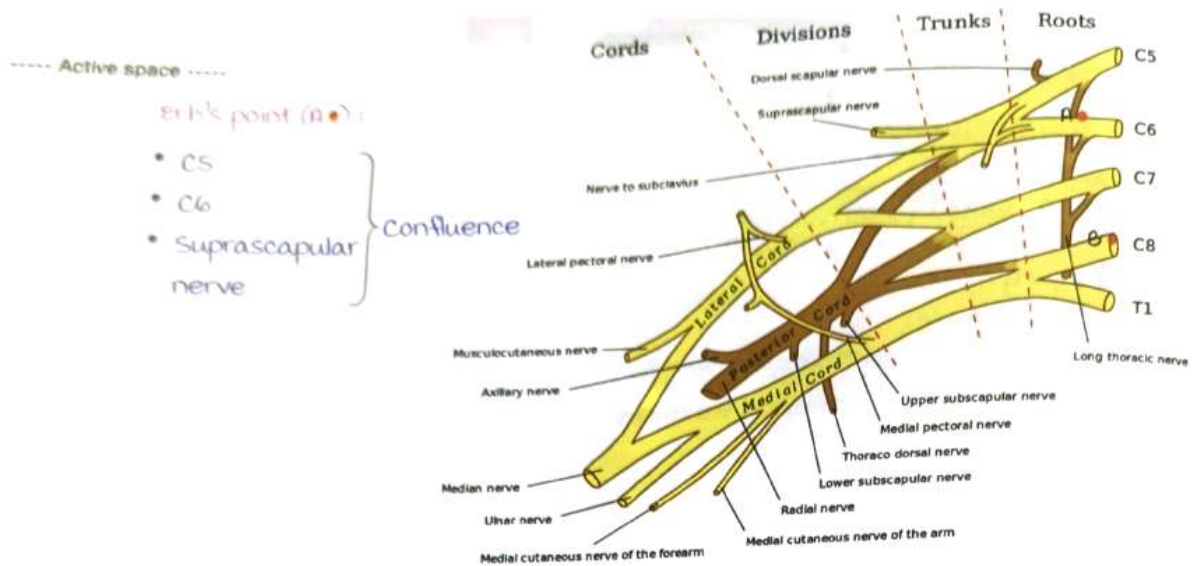
- Supraspinatus : 0-15°.
- Deltoid : 15-90°.
- Serratus anterior & trapezius : Overhead abduction.

**MUSCULOCUTANEOUS NERVE**

	motor supply	Sensory supply
Innervation	Anterior compartment of arm : • Biceps brachii : Forearm supination, elbow flexion. • Brachialis : Elbow flexion.	Lateral aspect of forearm
Deformity	Extended elbow with pronated forearm	Paraesthesia, tingling, numbness

BRACHIAL PLEXUS INJURIES**Anatomy of brachial plexus :**

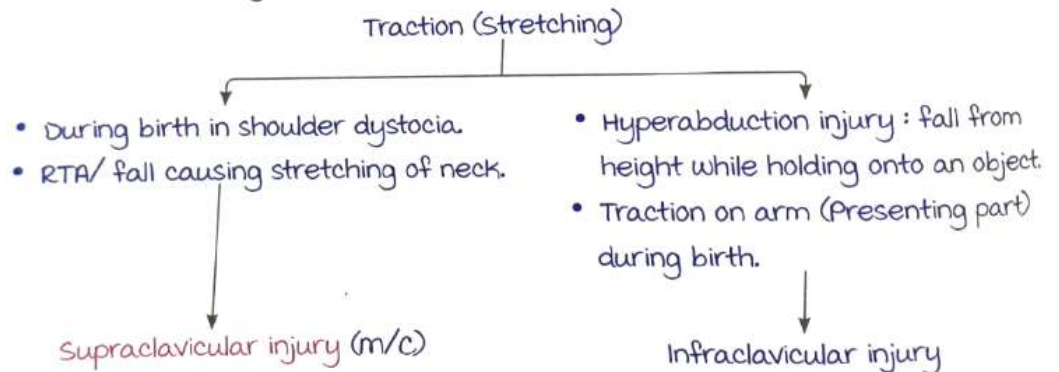
- Roots : C5, C6, C7, C8, T1.
- Trunks : upper, middle, lower.
- Divisions : Anterior & Posterior.
- Cords : Lateral, medial, posterior.



Types of brachial plexus injuries :

- Supraclavicular injury (upper roots & trunks) → Erb's palsy (Best prognosis).
- Infraclavicular injury (Lower roots & trunks) → Klumpke's palsy.
- Combined/mixed palsy (Worst prognosis).

Mechanism of injury :



Long thoracic nerve of bell :

- Root value : C5, C6, C7.
- Muscle innervated : Serratus anterior.
- If injured : On pushing against wall → Winging of scapula (Scapula lifts up).



Winging of scapula

Suprascapular nerve :

- motor supply — {
 - Supraspinatus : Shoulder abduction
 - Infraspinatus : External rotation of shoulder
- Deformity on injury : Adduction & internal rotation of shoulder.

erb's palsy :

Nerves injured

1. Suprascapular : Adduction, internal rotation.
2. Axillary : Adduction, internal rotation.
3. musculocutaneous : Elbow extension, forearm pronation.

Deformity

Policeman's/waiter's/
Porter's tip deformity

----- Active space -----



erb's palsy

klumpke's palsy :

C8,T1 injury

Combined ulnar + median nerve palsy

Complete clawing of hand

Loss of sympathetic supply to eye (T1)

Parasympathetic overactivity on eye

Horner's syndrome

(Ptosis, miosis, anhidrosis, loss of ciliospinal reflex)

Treatment of brachial plexus injury :

Aeroplane splint.



Aeroplane splint

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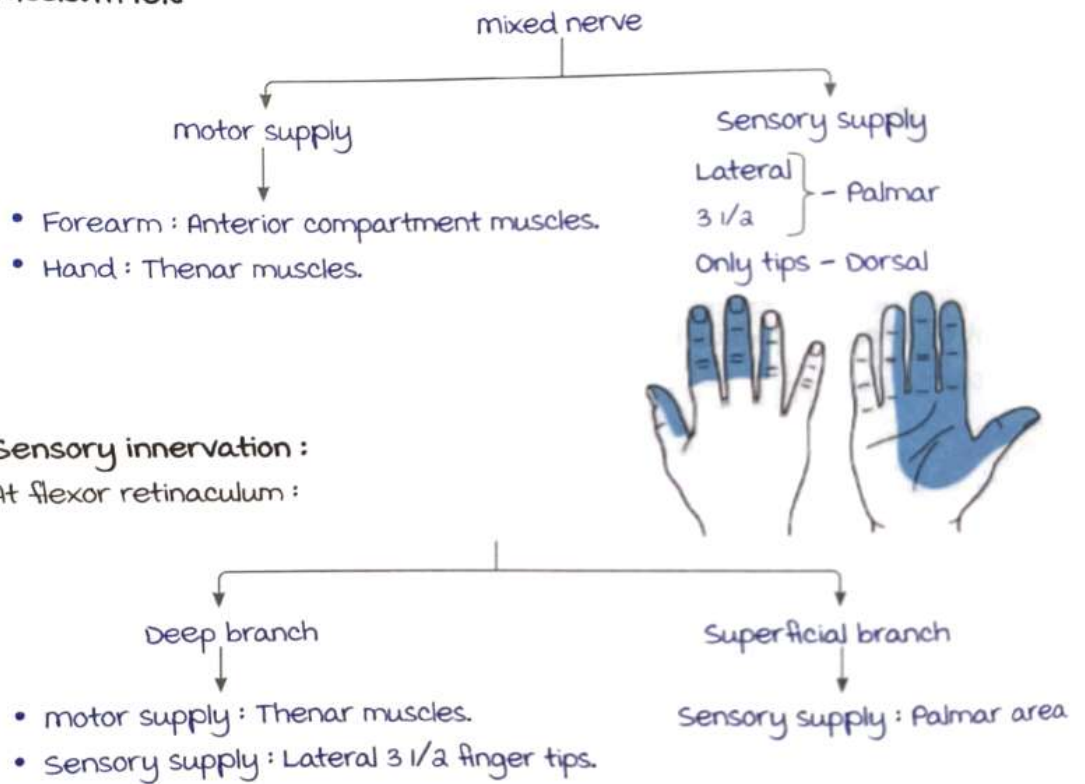
NERVE INJURIES : PART 2

Median Nerve

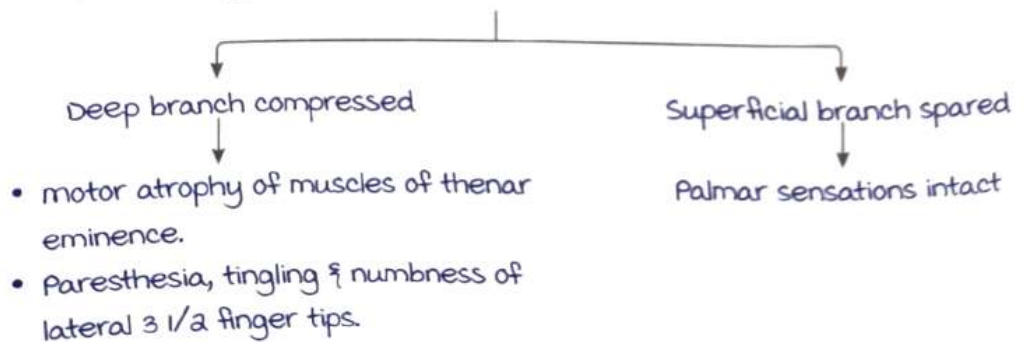
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AKA labourer's nerve.

INNERVATION



In carpal tunnel syndrome :

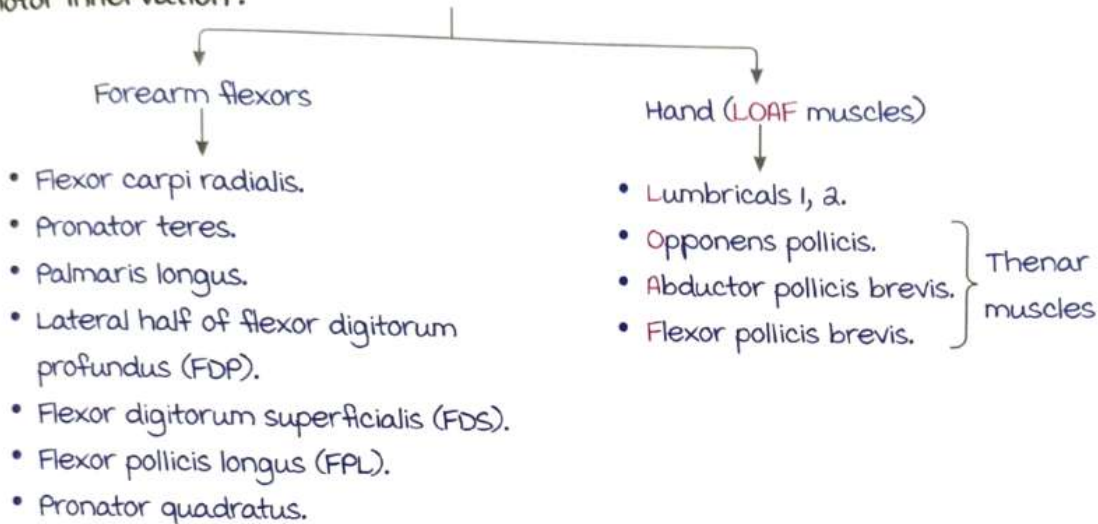


Autonomous zone :
Independent zone supplied by a single nerve.

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Nerve	Autonomous zone
median nerve	Tip of index finger
ulnar nerve	Tip of little finger
Radial nerve	Dorsum of first web space

motor innervation :



muscles innervated by median nerve :

Arm : No supply.

Forearm : All muscles of anterior (Flexor) compartment except
 Hand : All thenar muscles except adductor pollicis (Ulnar nerve)
 +
 lumbricals 1,2.

Flexor carpi ulnaris }
 medial half of FDP } supplied by ulnar nerve

MEDIAN NERVE INJURIES

Etiology — At wrist : Carpal tunnel syndrome
 — At elbow : Supracondylar humerus fracture

Anterior interosseous nerve (Branch of median nerve) >

median nerve > Radial nerve > Ulnar nerve

Median nerve injury at wrist :

manifestations :

• Loss of sensation of lateral 3^{1/2} fingers.

• Wasting of thenar eminence

• Loss of thumb — Abduction
 — Flexion
 — Opposition

Ape hand/thumb deformity



----- Active space ----- Treatment : **Opponens splint.**

Median nerve injury at elbow :

manifestations of wrist lesion + loss of functions of

→ FPL
→ FDS
→ FDP

Evaluation :

1. Pen test :

- Evaluates function of **adductor pollicis brevis** (Abduction of thumb).

• Procedure :

- Patient's hand placed on table palm facing up.
- Patient asked to touch tip of pen hovering above palm with tip of thumb



Unable to touch pen (Positive pen test)

2. Pointing index/Benediction sign/Pope sign/Oschner clasp sign :

Evaluates function of

→ FDS : Flexion of PIP & MCP joints

→ FDP : Flexion of DIP, PIP, MCP joints



Index finger doesn't flex on making a fist in median nerve injury at wrist

Note : Actions of thumb

- Abduction : Perpendicular to plane of palm.
- Extension : Parallel to plane of palm.

ANTERIOR INTEROSSEOUS NERVE

- Branch of median nerve.
- Purely motor nerve : Supplies
 - Flexor pollicis longus
 - Lateral half of FDP

m/c injured nerve in **supracondylar humerus fracture**

	AIN palsy	Median nerve palsy
Sensory deficits	-	+
Difficulty making fist	-	+

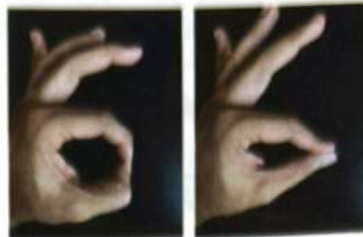


Evaluation :

Kiloh-Nevin sign : weak OK sign.

Note :

Lunate dislocation : Causes compression injury of median nerve.



Normal OK sign Weak OK sign

----- Active space -----

Summary on median nerve assessment :

1. median nerve → Pen test : For adductor pollicis brevis
→ Pointing index : For FDP & FDS
2. Anterior Interosseous nerve → Kiloh-Nevin sign (weak OK sign).

Ulnar Nerve

00:17:37

AKA musician's nerve.

Root value : C8, T1.

INNERVATION

motor supply :

Arm : No supply.

Forearm :

- medial half of FDP.
- Flexor carpi ulnaris.

Hand :

- Hypothenar muscles.
- Adductor pollicis (Thenar muscle) → Thumb adduction.
- Interossei group of muscles → Palmar → Adduction (PAD) of fingers
→ Dorsal → Abduction (DAB) of fingers
- Lumbricals 3,4.

Sensory supply : medial 1/2 fingers anteriorly & posteriorly.

Autonomous zone : Tip of little finger.

SITES OF ULNAR NERVE INJURIES

1. Cubital tunnel (Behind medial epicondyle).
2. Guyon's canal (Under piso-hamate ligament).

Ulnar nerve injury at wrist (In Guyon's canal) :

Manifestations :

- Partial clawing of hand.
- Hypothenar wasting.
- Loss of sensations in medial 1/2 hand.

Note :

Complete clawing : median + ulnar nerve palsy

----- Active space -----

Treatment : Knuckle bender splint



Knuckle bender splint



Partial claw hand



Complete claw hand

Evaluation :

1. Card test :



Card test

- To evaluate : **Palmar interossei** (Adduction of fingers).
- Procedure : Insert card between two extended fingers

↓
Examiner tries to pull card out from patient's fingers

↓ Weak interossei muscles

Card slips out easily (Positive card test)

2. Egawa test :

- To evaluate : **Dorsal interossei** (Abduction of 2nd, 3rd, 4th, 5th fingers).
- Method : moving middle finger away from midline.

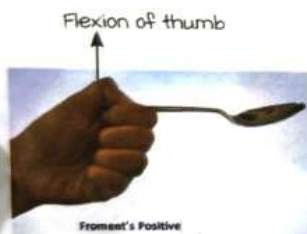
3. Fanning fingers against resistance : To evaluate **dorsal interossei**.

4. Book test :

- To evaluate : **Adductor pollicis**.
- Procedure : Patient asked to hold an object between thumb & index finger

↓
Lack of adduction compensated by flexors (AIN supply +)

↓
Flexion of thumb (Froment sign)



Froment's Positive

ulnar nerve injury at elbow :

Site of injury : In cubital tunnel, behind medial condyle

manifestations :

Loss of function of medial half of flexor FDP

↓
Loss of flexion at DIP, PIP and MCP joints

↓
Less significant clawing

ulnar paradox : Higher the injury of ulnar nerve → Lesser the clawing/deformity.

Summary on ulnar nerve assessment :

1. For Interossei :
 - Palmar : Card test
 - Dorsal :
 - Egawa test
 - Fanning of fingers
2. For Adductor pollicis : Book test.

----- Active space -----

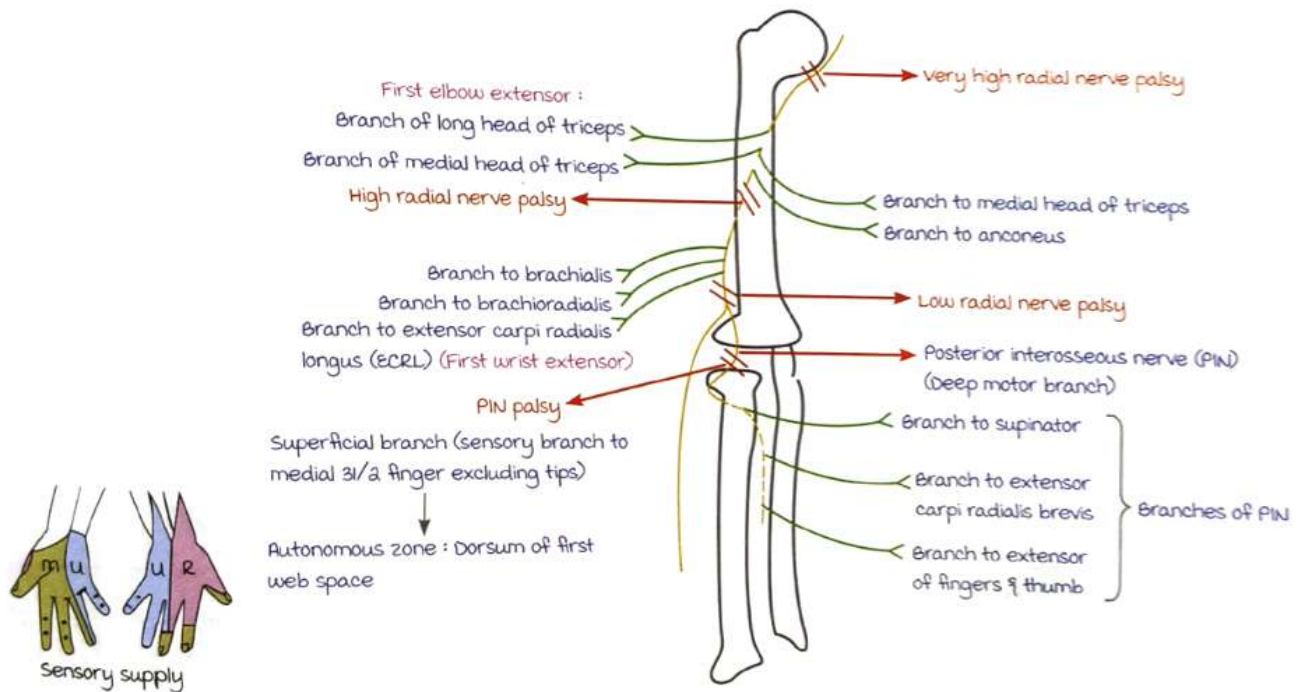
Radial Nerve

00:35:59

m/c injured nerve but best prognosis (Purely motor nerve).
Function : Extension of elbow, wrist & fingers.

Course & branches of radial nerve :

Origin : Posterior cord of brachial plexus.



Level of injury	Location of injury	manifestations
PIN palsy	Radial head	Finger drop, thumb drop
Low radial nerve palsy	Lateral condyle	Finger drop, thumb drop, sensory loss
High radial nerve palsy (m/c)	Spiral groove	Finger drop, thumb drop, sensory loss, wrist drop
Very high radial nerve palsy	Axilla	Finger drop, thumb drop, sensory loss, wrist drop, inability to extend elbow

Injury — Proximal to ECRL (Spiral groove) : wrist drop +
 — Distal to ECRL (Spiral groove) : wrist drop -

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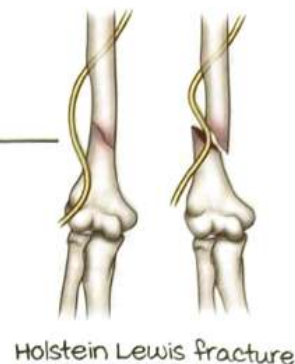
Holstein Lewis fracture :

m/c cause of radial nerve injury.

Fracture of shaft of humerus between
upper 1/3rd & lower 1/3rd of humerus

Causes radial nerve entrapment

Wrist drop, finger drop, thumb drop, sensory loss



Holstein Lewis fracture

Treatment :



Cock up splint

Cock up splint (Rx for wrist drop)

Static

(Extends wrist)

Dynamic

(Extends wrist + allows flexion of fingers)



Wrist drop

SUMMARY OF NERVE FUNCTION TESTS

Nerve	Test/sign	muscle
ulnar	Book test	Adductor pollicis
	Froment sign	Flexor pollicis substitutes for adductor pollicis
	Card test	Palmar interossei
	Egawa test	Dorsal interossei

Nerve	Test/sign	muscle
median	Ape thumb	Thenar muscle wasting
	Pen test	Abductor pollicis brevis
	Kiloh nevin sign/pincer grasp	Flexor digitorum profundus + flexor pollicis longus (AIN)
	Pointing index/benediction sign/oschsner clasp	Flexor digitorum superficialis + lateral half of flexor digitorum profundus
Radial	Finger & thumb drop	Extensors (PIN)
	Wrist drop	Extensors of wrist ECRL

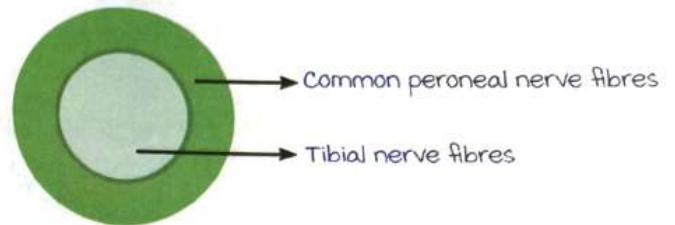
Lower Limb Injuries

sciatic nerve :

If injured causes manifestations of CPN nerve injury.

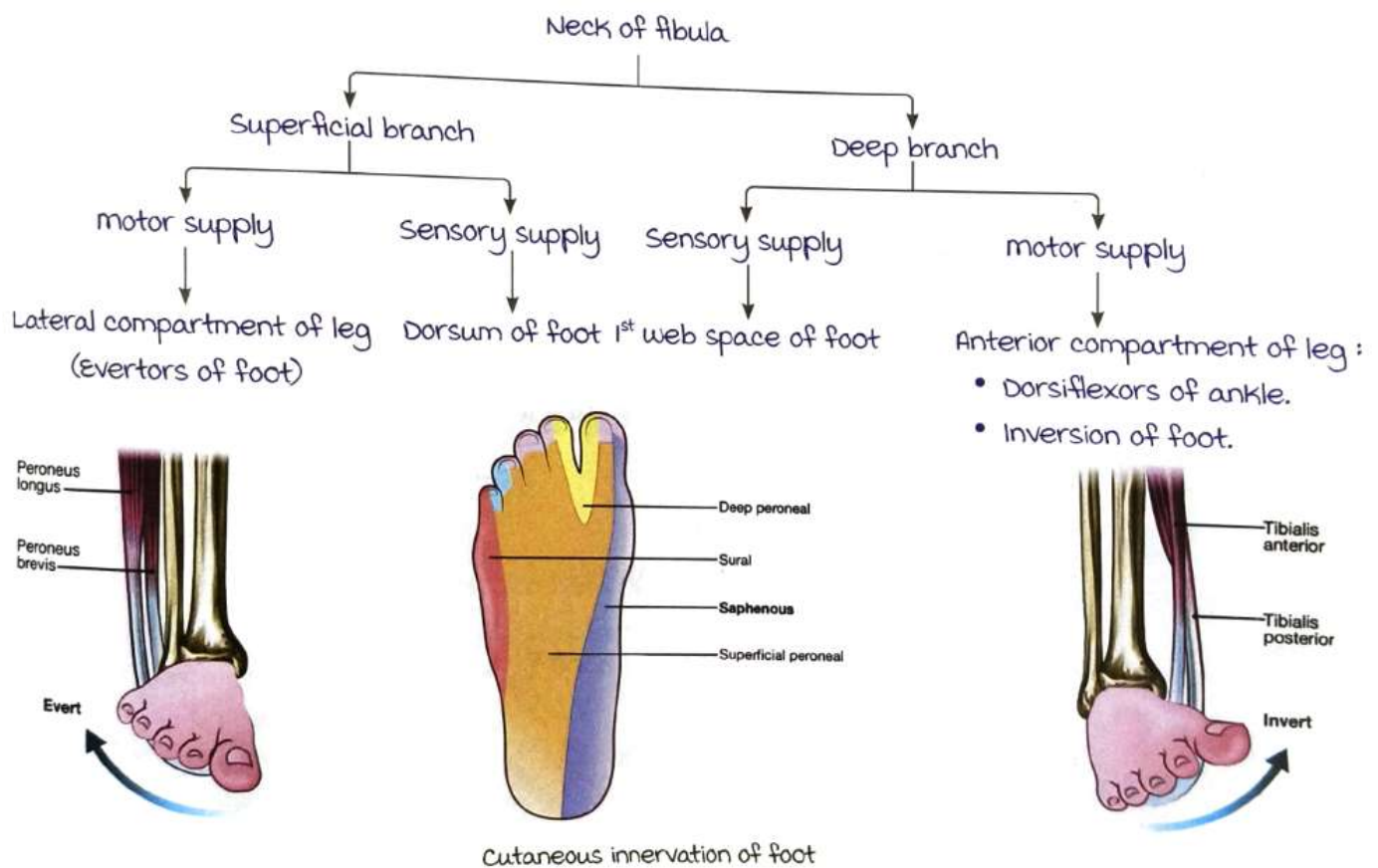
causes of injury :

- Iatrogenic (Hip surgeries) : m/c cause.
- Posterior dislocation of hip.
- Gunshot injuries.
- Gluteal injections.



Sciatic nerve : Cross section

Common peroneal nerve :



Causes of injury: Fracture of neck of fibula

manifestations:

- Loss of ankle — { Dorsiflexion } Equinovarus deformity
 { Eversion } (Plantar flexion & inversion)
- Gait : High stepping/foot-drop gait/steppage gait
(to overcome dragging of foot d/t foot drop)

----- Active space ----- Splint : Foot drop splint/Ankle Foot Orthosis (AFO).



Ankle foot orthosis



Foot drop



Fracture neck of fibula

Entrapment Neuropathies

00:57:05

Nerve trapped under a structure → Delayed conduction of stimulus.

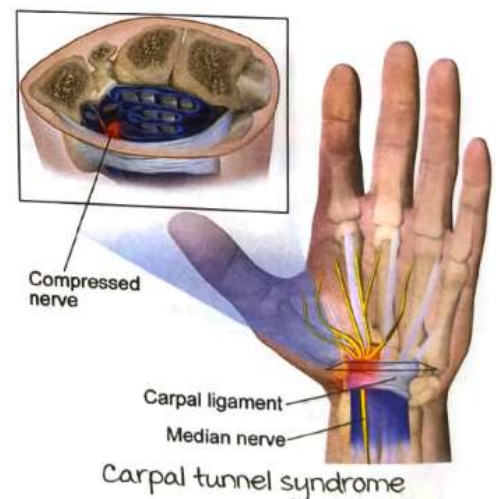
CARPAL TUNNEL SYNDROME

median nerve entrapment in carpal tunnel (at wrist) under flexor retinaculum/transverse carpal ligament.

m/c entrapment neuropathy (m/c entrapped nerve : median nerve).

Etiology (Conditions causing ↑ pressure inside carpal tunnel) :

- Idiopathic : m/c cause.
- Hypothyroidism (Myxedema).
- Rheumatoid arthritis (Inflammation).
- Pregnancy (Fluid retention).
- Acromegaly (Excessive growth of bone & soft tissue).
- Activity causing wrist compression (Eg : typing).
- Colles' fracture (Hematoma → Compression).
- Gout.
- Amyloidosis.
- Diabetes mellitus.



Clinical features :

- middle aged female.

- c/o pain :

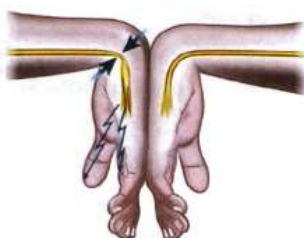
- Type of pain : **Burning pins & needle** type of pain.

- Distribution : Along lateral 3 1/2 fingers — **maximum at night**
 Relieved on — Shaking her hand
 Hanging hand at the edge of bed

- Thenar muscle wasting & weakness.

Evaluation :**Clinical tests :****1. Phalen's test**

Position held for
approx. 1 minute



Dull aching pain, numbness,
paraesthesia, tingling along
median nerve distribution

2. Reverse Phalen's test

Position held for
approx. 1 minute

3. Durkan's test :

- Best clinical test.
- Direct median nerve compression between thenar & hypothenar eminence for 30 seconds → Reproduces symptoms.

4. Tourniquet test.**Nerve conduction studies :**

- **100.**
- Determines nerve conduction velocity.
- **Decreased velocity** d/t nerve compression under flexor retinaculum.

Treatment :

Conservative : Rest, steroids, splints

↓ No improvement

Surgery : Release of flexor retinaculum/transverse carpal ligament.

Note :

Allen's test : To assess patency of vasculature of hand before performing arterial procedures (Arterial line).

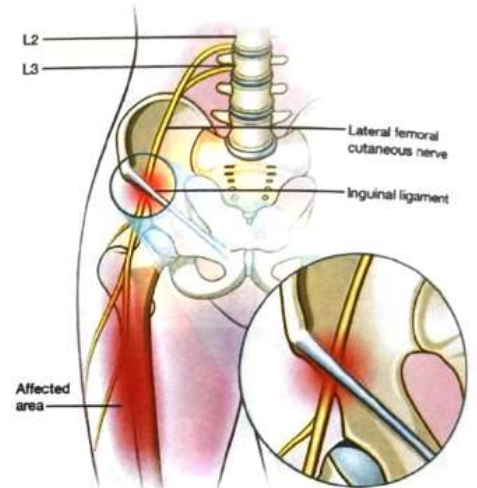
----- Active space ----- Opposition of thumb :

- muscle involved : Opponens pollicis → Supplied by deep branch of median nerve.
- Only function completely lost in CTS.

MERALGIA PARAESTHETICA

- Entrapment of lateral cutaneous nerve of thigh under inguinal ligament.
- Burning sensation along lateral aspect of thigh.

Etiology — { Overweight
Tight clothing around hip



The nerve is compressed by the inguinal ligament.

ENTRAPMENT NEUROPATHIES

Syndrome/pathology	Nerve	Site of compression/injury
Carpal tunnel syndrome (m/c)	median	At wrist
Cubital tunnel	ulnar	Behind the medial condyle humerus
Guyon's canal	ulnar	under the Pisio-hamate ligament
Pronator syndrome	median nerve	Heads of pronator teres
Kiloh Nevin syndrome	AIN	Supracondylar humerus fracture
Cheralgia paraesthetica	Superficial radial nerve	Radial styloid, insertion of brachioradialis (wrist watch, cuffs, bangles)
meralgia paraesthetica	Lateral cutaneous nerve of thigh	under inguinal ligament (tight belt)
Pyriformis syndrome	Sciatic nerve	Pyriformis & obturator internus muscle
Tarsal tunnel syndrome	Posterior tibial nerve	Behind medial malleolus below the flexor retinaculum of foot (RA)
morton metatarsalgia	Interdigital nerve of foot	Between 3 rd & 4 th toe

ORTHOPAEDIC ONCOLOGY : PART 1

----- Active space -----

Classification of Bone Tumors

00:01:46

Enneking classification :

Father of orthopaedic oncology : William Enneking.

Benign bone tumor	Malignant bone tumor
1. Latent : Thick capsule Eg : Non ossifying fibroma	I : Low grade → A : Intra compartmental (Within bone) → B : Extra compartmental (Outside bone)
2. Active : Thin capsule Eg : Aneurysmal bone cyst	II : High grade → A : Intra compartmental → B : Extra compartmental
3. Aggressive : Capsule is breached Eg : Giant cell tumor	III : metastasis

Classification based on cell type :

Cell type	Benign	Malignant
Chondrocytes (Cartilage forming cell)	- Osteochondroma - Enchondroma - Chondroblastoma	Chondrosarcoma
Osteoblast (Bone forming cell)	- Osteoid osteoma - Osteoblastoma	Osteosarcoma
Osteoclasts (Giant cell rich tumor)	Osteoclastoma/Giant cell tumor	
	- Aneurysmal bone cyst - Non-ossifying fibroma	

Appearance on X-ray :

Cell type	Appearance
Chondrocytes	Lucent/hazy with calcification
Osteoblasts	Dense/sclerosed
Osteoclasts	Bone destruction

Miscellaneous bone tumors :

Tumor	Type	Features
Hemangioma	Benign	- Vascular tumor - Occurs in spine
Simple bone cyst	Benign	Mesenchymal tumor of bone
Fibrous dysplasia	Benign	- Mesenchymal tumor of bone - Bone is replaced by fibrous tissue
Ewing's sarcoma	Malignant	- Small round blue cell tumor family - Arises from medullary cavity of bone

----- Active space ----- Note : Osteoid osteoma produces maximum bone matrix.

Classification based on location in the bone :

Location	Tumors
Epiphysis	- Chondroblastoma : Growth plate visible (Child). - Giant cell tumor : No growth plate (Adult).
Diaphysis	- Ewings sarcoma - Osteoid osteoma (Night pains responding to aspirin). - Adamantinoma.
metaphysis	- Osteosarcoma. - Osteochondroma. - Aneurysmal bone cyst. - unicameral bone cyst. - Non-ossifying fibroma. - Chondromyxoid fibroma.

Classification based on nature of tumor :

Benign	Benign tumors with malignant potential	Malignant
- Osteoid osteoma - Simple bone cyst	- Giant cell tumor - Osteoblastoma - Osteochondroma	- Chondrosarcoma. - Osteosarcoma. - Ewings sarcoma. - Adamantinoma. - Chordoma.

Classification based on age groups :

Age	Tumor	Remarks
5-25 years	Ewings sarcoma	- m/c malignant tumor in the 1st decade of life. - m/c occurs in 2nd decade of life.
10-20 years	Primary osteosarcoma	m/c malignant bone tumor in children
	Secondary osteosarcoma (40-60 years)	
20-40 years	Giant cell tumor/osteoclastoma	D/t Paget's disease or radiation exposure
40-60 years	Secondary osteosarcoma Chondrosarcoma	
>60 years	metastasis, multiple myeloma Adamantinoma, Chordoma	

Miscellaneous Topics

00:20:40

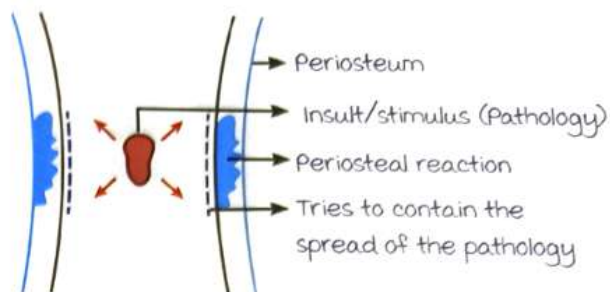
----- Active space -----

periosteum

- Reaction of periosteum to an insult/stimulus.
- Periosteal insult → New bone formation → Prevents the spread of underlying pathology.

Types :

- Slow growing/indolent lesion : Thick/solid periosteal reaction.
- Rapidly growing/aggressive lesion : Thin/lamellated periosteal reaction.



Reaction of periosteum to an insult/stimulus

x-ray features of periosteal reaction :

Features	Tumors
Onion peel	Ewings sarcoma
Sunburst	Osteosarcoma
Codman's triangle	Osteosarcoma
Solid	Osteomyelitis



Codmans triangle



Lamellated/layered

Sunburst appearance
: D/t calcification of
Sharpey's fibres

Pulsatile bone tumors

Vascular → Hence they are pulsatile → Bruit present.

Eg :

- Osteosarcoma → most pulsatile.
- Aneurysmal bone cyst.
- Giant cell tumor.
- METS from kidney.
- METS from thyroid.

----- Active space -----

Polyostotic lesions : multicentric

Can occur in multiple parts of same bone or in multiple bones (Not metastatic).

- Fibrous dysplasia.
- Giant cell tumor.
- Enchondroma.
- Osteochondroma.

Fibrous cortical defect/non-ossifying fibroma

- Location : Cortex.
- Defect is filled with fibrous tissue.
- Eccentric lesion (Away from centre).
- Self-resolving lesion.
- Occurs in metaphysis.
- Occurs in 1st decade of life.

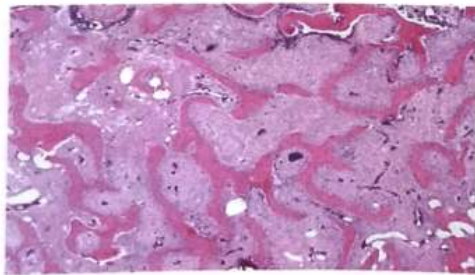
**Fibrous dysplasia****Pathology :**

- Developmental anomaly of the bone.
- Bone tissue replaced by fibrous tissue.
- m/c occurs in femur.

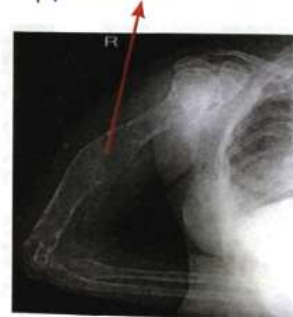
Types $\begin{cases} \text{monostotic (m/c)} \\ \text{Polyostotic} \end{cases}$

Biopsy (Best ix) :

Chinese letter pattern appearance.

**X-Ray :**

1. Ground glass appearance



2. RIND sign



3. Shepherd crook deformity :

**Syndromes associated with fibrous dysplasia :**

1. McCune-Albright syndrome :

- Polyostotic fibrous dysplasia (multicentric).
- Pigmentation : Café au lait spots.
- Precocious puberty.

2. Mazabraud syndrome :

- Polyostotic fibrous dysplasia.
- Intramural (muscular) myxomas.

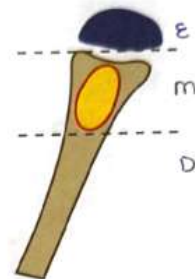
Bone Cyst

00:34:51

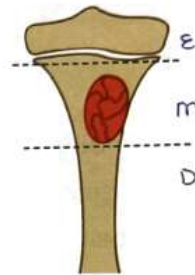
----- Active space -----

simple bone cyst (SBC) v/s Aneurysmal bone cyst (ABC) :

simple bone cyst (SBC)	Aneurysmal bone cyst (ABC)
unilocular	multi-loculated
symmetrical expansion	Assymetric expansile/ballooning
centric	Eccentric
metaphyseal	
10-20 years old	
Proximal humerus	Around the knee
Contains clear/straw coloured fluid	Contains blood
Ex:	
- Aspiration ± injection of steroids/sclerosants - Excision & curettage with autologous bone graft	Extended curettage using liquid nitrogen, phenol or bone cement



Simple bone cyst



Aneurysmal bone cyst

Note : Extended curettage is also done in giant cell tumor.

Radiological features of simple bone cyst :



Simple bone cyst

Growth plate (+)
(child)

metaphyseal lesion



Fallen leaf sign

Chip of the cortex breaks
& freely falls down

Trap door sign

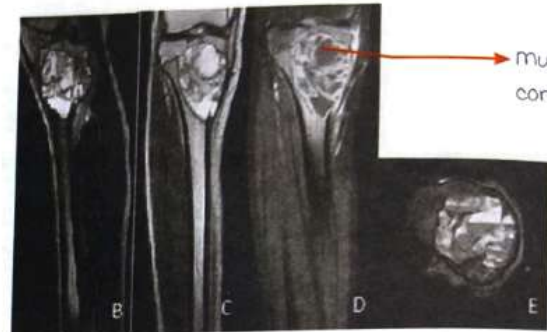
Chip of bone hinges
at the cortex

----- Active space ----- Radiological features of aneurysmal bone cyst :



Aneurysmal bone cyst

- Growth plate (+)
(Indicates its a child)
- multiloculated lesion in the metaphysis



MRI of aneurysmal bone cyst

multiple fluid containing levels

Note :

1. D/D for multiple fluid containing levels :

- Aneurysmal bone cyst.
- Giant cell tumor.
- Telangiectatic osteosarcoma.

2. D/D for cystic lesion of bone :

Centric	Eccentric
• Simple bone cyst • Enchondroma • Brodie's abscess	• Giant cell tumor • Aneurysmal bone cyst • Non ossifying fibroma

ORTHOPAEDIC ONCOLOGY : PART 2

----- Active space -----

cartilaginous lesion & calcifications :

- chondrocyte predominant.
- popcorn calcification/Ring and arc calcification/O ring sign :
 - chondroma.
 - osteochondroma.
 - chondrosarcoma.



Popcorn calcifications

Chondrocyte Predominant Tumors

00:01:04

OSTEOCHONDROMA/EXOSTOSIS

General features :

- m/c benign bone tumor.
- m/c site : Distal femur.
- Presents as mass around the joint.
- sessile/pedunculated.
- Developmental malformation of growth plate : Aberrant development.
- Follows normal direction of growth : Epiphysis to metaphysis (Away from the joint).
- Stops growing after skeletal maturity (Not a true tumor).

Clinical features :

- Asymptomatic, grows with skeleton.
- Stops growing after skeletal maturity.
- Pain.
 - Bursitis (m/c cause).
 - Nerve compression.
 - Fracture.
- malignant transformation → Chondrosarcoma.

Signs :

- >2 cm cap (On MRI).
- Heavy calcification of cap (On X-ray).
- Persistence of growth after skeletal maturity.



X-ray osteochondroma

----- Active space -----

management :**Radiological features :**

- Large to feel but small on X-ray.
- Cartilaginous cap (<2 cm) cannot be visualized.

Architecture of the lesion matches normal bone

**Treatment :**

Extraperiosteal excision after skeletal maturity.

Hereditary/multiple exostosis/diaphyseal aclasis :

- Autosomal dominant inheritance.
- EXT 1, 2 gene mutation.
- 5% cases → Chondrosarcoma.



multiple exostosis

ENCHONDROMA**General features :**

- m/c bone tumor in small bones of hands & feet.
- Location : metaphyseal.
- Rx : Extended curettage + bone graft.

Lucent lesion in metacarpal (Cartilaginous)



Enchondroma

Associated syndromes :**1. Ollier's syndrome**

- multiple enchondromas.
- 30% cases are premalignant.
- very rare.



Ollier's syndrome

2. Maffucci's syndrome

- multiple enchondromas.
- Cavernous hemangiomas, lymphangiomas.



multiple hemangiomas of hand

Note :

- m/c tumor of the hand : Squamous cell carcinoma.
- D/D for enchondroma : Spina ventosa/TB dactylitis.
 - variant of TB involving fingers.

CHONDROBLASTOMA

- AKA *Codman's tumor*.

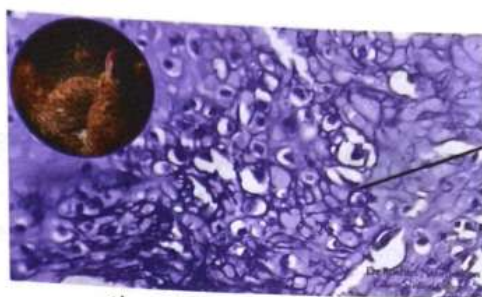
- Epiphyseal lesion.

- m/c in children.

- 10-25 yrs (Before skeletal maturity).

X-ray : Punctate/stippled calcification.

Treatment : Excision curettage with autologous bone graft.



Chondroblastoma : Biopsy

Osteoblast Tumors

00:13:54

OSTEOID OSTEOMA

- m/c *true benign tumor*.

- Children and adolescents.

- Location : Diaphysis.

- m/c site : *Femur*.

- Eccentric/cortical lesion.

Clinical features :

- Central nidus → Produces prostaglandins → Pain.
- Night pain that responds to aspirin.
- Swelling (In thigh or leg).

X-ray :

Central lucent nidus with dense sclerosis around

Osteoid osteoma

Treatment :

- NSAIDs (Pain relief).
- Radiofrequency ablation.
- Excision curettage.
- MR guided focused ultrasound destruction.

OSTEOBLASTOMA

- Central nidus > 2 cm.

- Occurs in spine.

- May not respond to aspirin.

- Risk of malignant transformation → *Osteosarcoma*.

Treatment :

- Surgical excision.



Osteoblastoma

----- Active space -----

Giant Cell Tumor/Osteoclastoma

General features :

- Locally aggressive.
- Location : Epiphysio-metaphyseal.
- Females > males.
- Age : 20-40 years (After skeletal maturity).
- Site :
 - Distal end of femur (m/c).
 - Proximal end of tibia.
 - Distal end of radius (Classical).
- m/c tumor occurring in distal end of radius : Osteoclastoma.



Osteoclastoma/Giant cell tumor

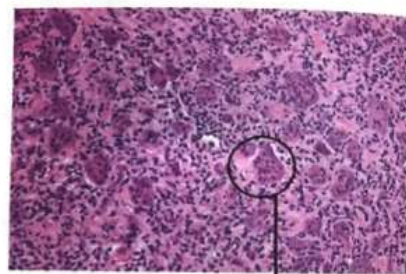
Clinical features :

Eggshell crackling (Thin cortex breaks on palpation).

X-ray : Soap bubble appearance (Left over bony trabeculae after destruction).

Biopsy :

- multinucleated osteoclast like giant cells, surrounded by mononuclear cells (Tumor cells).
- Tumor cells → ↑ RANK ligand production.



Osteoclastoma : multinucleated osteoclast like giant cells

Treatment :

- Extended curettage with liquid nitrogen, phenol or bone cement (To destroy residual cells).
- Followed by autologous bone graft.

Mnemonic : E

- Epiphyseal location.
- Occurs after epiphyseal fusion.
- Eccentric.
- Expansile.
- Eggshell crackling.
- Extended curettage.

Giant cell tumor variants :

Tumors rich in giant cells :

- Non-ossifying fibroma (m/c).
- Aneurysmal bone cyst (Closest resemblance).
- Chondromyxoid fibroma.
- Chondroblastoma.
- Fibrous dysplasia.
- Simple bone cyst.
- Brown's tumor.
- Osteosarcoma (Telangiectatic).

----- Active space -----

Note :

1. <5% cases : Risk of malignant transformation.

- 2° osteosarcoma.

- Fibrosarcoma.

- malignant fibrous histiocytoma (Pleomorphic sarcoma).

} 3% metastasise to lungs.

2. 20% cases : Local recurrence.

Hemangioma

00:29:43

General features :

- Occurs in elderly population.
- Benign, asymptomatic vascular bone tumor.
- Site : Spine > Skull > Pelvis.

Radiological features :

X-ray :

- Vertical striations : Jail bar/jail house appearance.
- Corduroy appearance.



Jail bar/jail house appearance

CT scan :

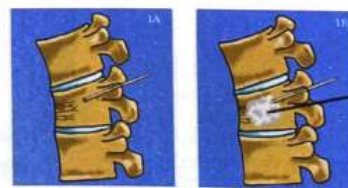
Polka dot sign (Axial view).



Polka dot sign

Treatment :

- Conservative.
- Radiotherapy.
- Vertebroplasty.



vertebroplasty

Injecting bone cement

Chondrosarcoma

00:31:04

General features :

- Location : metaphysis.
- Age : 40-50 yrs.
- Site : Pelvis, proximal femur (Calcified tumor).
- A/w paraneoplastic syndrome : Hyperglycemia.

X-ray :

Endosteal scalloping and punctate calcification.



Chondrosarcoma

----- Active space -----

Osteosarcoma

00:32:43

General features :

- m/c bone tumor seen in children.
- Bimodal presentation
 - 2nd decade : 1° osteosarcoma.
 - Older population : 2° osteosarcoma.
(D/t paget's disease or radiation exposure)
- 1° osteosarcoma > 2° osteosarcoma.
- Radiation-induced osteosarcoma usually occurs 10 yrs after exposure.
- Location :
 - metaphysis.
 - Distal end of femur.
- most radio-resistant tumor.
- m/c radiation induced bone tumor.
- Bone forming tumor : Forms abnormal osteoid and matrix.
- 15% cases : micrometastasis to lungs during presentation
 - ↓
 - Pneumothorax.

X-ray :

Periosteal reaction



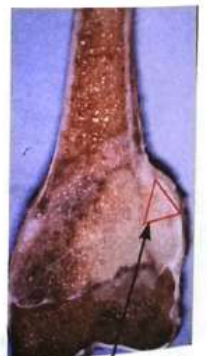
Codman's triangle



Sunburst appearance

Gross specimen :

Calcified Sharpey's fibres



Codman's triangle

Types :

1. Classical/intermedullary (m/c).
2. Periosteal.
3. Parosteal.
4. Telangiectactic.

Treatment :

Neoadjuvant chemotherapy (Prior to surgery)

↓
Surgery/limb ablation↓
Adjuvant chemotherapy (Post surgery)

chemotherapeutic agents : T10 protocol.

- Actinomycin.
- Cyclophosphamide.
- Bleomycin.
- High dose methotrexate.
- Doxorubicin.
- Vincristine.

----- Active space -----

Ewings Sarcoma

00:38:18

General features :

- m/c tumor in 1st decade of life.
- Occurrence : m/c in 2nd decade (5-20 yrs).
- Location : Diaphysis of femur (mid thigh swelling).
- males > females.
- High grade sarcoma.

Poor prognostic features :

- metastasis.
- male.
- Fever, anemia.
- ↑ ESR.
- Age > 12 yrs.
- Leukocytosis.
- Chemoresistance.
- Relapse.
- Size of the lesion.

Clinical features :

- Presents like infection.
- mid thigh/leg swelling.
- Signs of inflammation.
- Incidental h/o trauma.

X-ray :

Onion peel/lamellated appearance.



Onion peel/lamellated appearance

Biopsy : Best Ix

- Small, round blue cells with pseudo-rosettes.
- PAS +ve & Diastase digestible.
- MIC-2 (CD99) on immunohistochemistry :
Specific marker.
- Karyotyping : Translocation $t(11;22)$ m/c.
Other translocations :
 - $t(2;13)$, $t(7;22)$.
 - Trisomy 18.
 - Trisomy 12.

Treatment :

- Radiosensitive (But radiotherapy is not preferred d/t ↑ chance of recurrence).
- Current trend : Chemotherapy + limb salvage/resection ± adjuvant radiation.

Note :

Primitive neuroectodermal tumors with small blue round cells on HPE :

- Neuroblastoma.
- Retinoblastoma.
- medulloblastoma.

Miscellaneous Tumors

00:43:49

Chordoma :

- Origin : Physaliferous cells of primitive notochord.
- m/c location : **Sacrum**, base of skull, vertebrae.
- Occurs in **elderly**.

Adamantinoma :

- Low grade **malignant epithelial tumor** (That grows into bone).
- m/c bone : **Tibia** (Subcutaneous bone).
- Location : Diaphysis.



Diaphyseal soap bubble appearance

Synovial sarcoma :

- Does not arise from synovium.
- Arises from capsule, bursae or tendon.
- D/t **t(x;18)** → **SYT : SSX fusion gene**.

Metastasis

00:45:58

- m/c malignant bone tumor : **metastasis**.
- Source : Breast > Prostate > Lungs.
- m/c go to : **Lumbar vertebrae** > Thoracic vertebrae.

Characteristics of secondaries :

- | | |
|------------------------------------|------------------------------------|
| 1. Purely blastic : | 2. Purely lytic : |
| • Prostate . | • Kidney (RCC) . |
| • medulloblastoma. | • Follicular carcinoma of thyroid. |
| • Carcinoids. | |
| 3. mixed : Breast (Or only lytic). | |



Winking owl sign

Bone to bone metastasis :

1. Ewing sarcoma.
2. Osteosarcoma.

Note :

1. m/c primary malignant bone tumor : **multiple myeloma** > Osteosarcoma.
2. m/c non hematological malignant bone tumor : **Osteosarcoma** > Chondrosarcoma.
3. m/c benign bone tumor : Osteochondroma.
4. m/c true benign bone tumor : Osteoid osteoma.

INFECTION OF BONE AND JOINTS

----- Active space -----

INFECTIONS OF SKELETAL SYSTEM

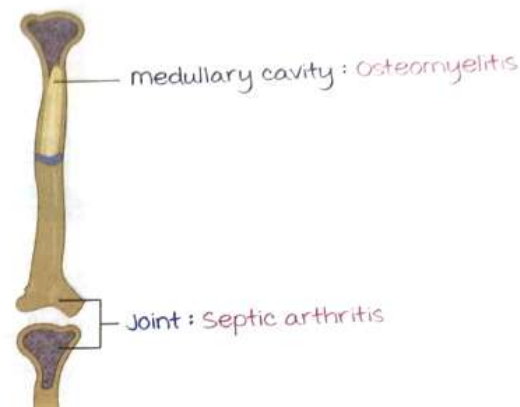
Based on site of infection :

Osteomyelitis :

- Bone & medullary cavity.
- m/c location : metaphysis of long bone.

Septic arthritis :

- Joint



Bone infections

Etiology :

Bacteria (m/c)

Pyogenic organisms :

Severe local/periosteal reaction

TB :

- Limited local reaction
(↓ inflammatory reaction)
- TB of short bones (Hand & foot) → Severe local reaction.

Osteomyelitis

00:03:08

In young children or elderly population : D/t relatively low immunity.

Routes/source of infection

- Hematogenous (m/c) : From pre-existing infection.
- Direct (Eg : Puncture wounds)
- Post surgical (iatrogenic)
- Open fracture.
- Septic arthritis : Joint Infection → Bone.

Etiology

Staphylococcus aureus :

- m/c organism infecting musculoskeletal system.
- m/c cause of :
 - Osteomyelitis overall.
 - Acute osteomyelitis.
 - Chronic osteomyelitis.
- m/c cause of osteomyelitis in :
 - Developed countries.
 - Developing countries.
 - Immunocompromised (HIV/AIDS).
 - Open fracture.
 - Post surgical.

----- Active space -----

Special cases :

Osteomyelitis in	Organism
Sickle cell disease patients	<i>Salmonella</i> → Diaphyseal involvement (If metaphysis involved : <i>Staphylococcus aureus</i>)
IV drug abusers	<i>Pseudomonas</i>
Following animal bite	<i>Pasteurella</i>
Following Human bite	<i>Eikenella</i>
Prolonged parenteral therapy	Fungal organisms

Site :

m/c :

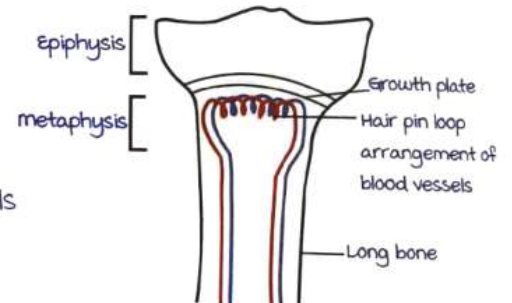
- Overall/infant/children : *Femur* (Distal > proximal) > *Tibia*.
- Adult : *vertebrae*.

Reasons of metaphyseal involvement :

- most vascular region of bone
(↑ nutrition for organism).
- Hair pin loop arrangement of blood vessels

↓
sluggish blood flow (Easy exit to bone).

- Few monocytes & macrophages
(↓ protection).

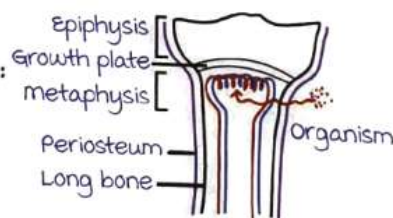


Pathophysiology & Clinical Features

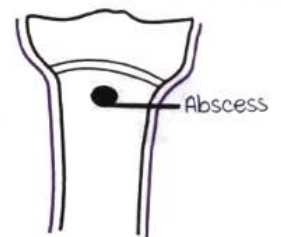
00:11:07

Pathophysiology :

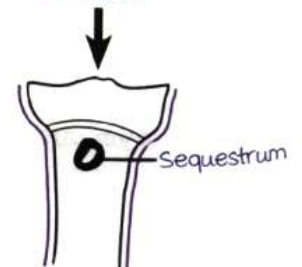
Acute osteomyelitis :



1. organism reach metaphysis : <24 hours
-Systemic manifestations : Fever, pain.

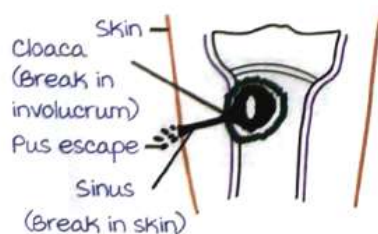


2. Abscess formed : After 24 hours.

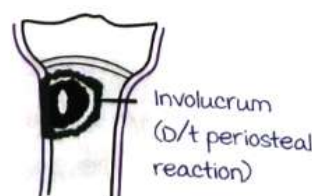


3. Sequestrum (Dead bone) formed

Chronic osteomyelitis :



5. Cloaca
↓
Sinus
- Pus drains out



4. Periosteal reaction : 7-10 days
↓
Involucrum (New bone that contain the spread of infection)

Clinical features :

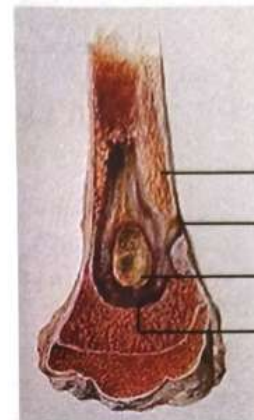
young child presents with :

- High grade fever.
- Pain & swelling.
- pseudo paralysis : Patient doesn't move limb d/t pain.

Note :

causes of pseudo paralysis

- Septic arthritis
- Osteomyelitis
- Scurvy (vit C deficiency) :
Subperiosteal hemorrhage → Pain.



Osteomyelitis

	Acute	Chronic
Clinical hallmarks	Not specific (Fever & pain)	Sinus
Pathological hallmarks	Abscess	Sequestrum

Management & Complications of Acute Osteomyelitis

00:23:38

INVESTIGATIONS

Blood investigations :

- Blood culture : 50% +ve.
 - WBC : Neutrophil
 - ESR
 - CRP
 - Serum procalcitonin.
- High, not specific.
- Sensitive & specific marker for
- Acute osteomyelitis.
 - Septic arthritis.

Radiology :

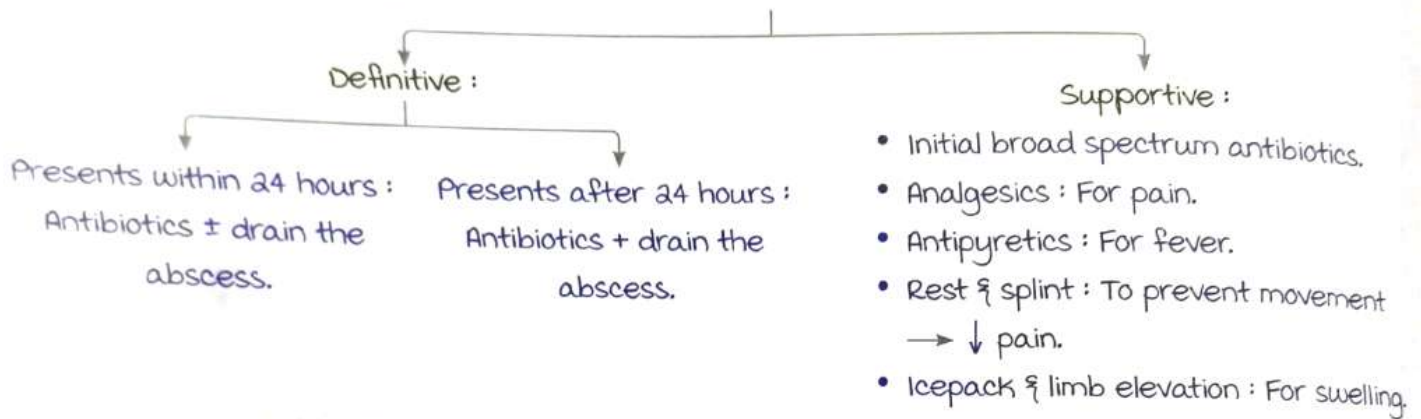
- MRJ :**
 - Picks up earliest changes within 24 hrs of disease onset : marrow edema.
- Bone scan :**
 - Technetium 99
 - Gallium 67
 - Indium 111 labelled WBC

Accumulation in infection
- X Ray :**
 - Changes detected after 24 hours of disease onset.
 - Earliest X-ray change (within 48 hours) : Soft tissue lucency/shadow around bone.
 - Earliest bony changes on X-ray (7-10 days/2 weeks) : Periosteal reaction..

----- Active space -----

Biopsy :

- Gold standard.
- Tissue sample collected from site of infection.

TREATMENT**Antibiotics :**

- 6 weeks : 2 weeks IV → 4 weeks PO.
- Long duration d/t low penetration of drug in bone.
- Response assessed using CRP measurements.

COMPLICATIONS

1. Chronic osteomyelitis : m/c.
2. Septic arthritis : Bone $\xrightarrow{\text{Infection}}$ Joint.
 - Prevented by growth plate.
 - Common in :
 - <2 yr : before growth plate formation.
 - Adults : After growth plate fusion.
3. Septicemia.
4. Growth disturbance.

Note : Duration

- Acute : <2 weeks
- Subacute : 2-4 weeks.
- Chronic : >4 weeks.

Chronic Osteomyelitis

00:33:27

- Sequelae/complication of acute osteomyelitis.
- >4 weeks following infection.
- Clinical hallmark : Sinus.



SEQUESTRUM

- Pathological hallmark of chronic osteomyelitis.
- Dead, dense, devascularized bone inside abscess.
- Surrounded by granulation tissue and involucrum.
- Microscopically: Destruction of haversian system.

----- Active space -----



Chronic osteomyelitis

Types of sequestrum :

Sequestrum	Associated conditions
Coraliform	Perthe's
Annular/ring	Amputation stump/pin tract sites
Tubular/diaphyseal	Acute pyogenic osteomyelitis
Feathery/flake/coarse sandy	Tuberculosis (TB)
Fine sandy	Viral osteomyelitis
Button hole	Post radiation
Ivory	Syphilis



Infection around pin tract

MANAGEMENT & COMPLICATIONS**Investigations :**

- Blood investigations : Culture, WBC, ESR, CRP.
- Radiology :
 - X-Ray : Sequestrum, involucrum, cloaca
- Sinogram : Dye injected through sinus → Anatomy of sinus tract seen → Plan Rx.
- Biopsy : Gold standard

Treatment :**Initial :**

- Broad spectrum antibiotics.
- Supportive treatment.

----- Active space ----- Definitive : Surgery.

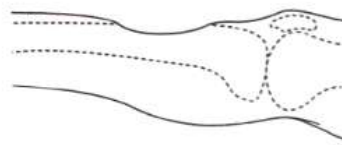
Antibiotic cover

Sequestrectomy : Removal of sequestrum.

Curettage : Till fresh bleed seen → Paprika sign



Debridement : Saucerization (wide mouth cavity made)



Closure of dead space with :

- Bone graft.
- Bone cement.



Skin covering with:

- Skin flap.
- muscle flap.
- Vacuum assisted covering (VAC)/Negative pressure wound therapy :
 - -75 to -120 mmHg
 - Promote granulation tissue formation
 - C/I : Eschar tissue or tumor.



VAC

Complications :

1. Pathological fracture (m/c).
2. Acute exacerbation.
3. Squamous cell carcinoma (SCC) : D/t Neoplastic changes of sinus tract.
4. Amyloidosis.

Variants of Osteomyelitis

00:44:13

----- Active space -----

No sinus or sequestrum.

Pathogenesis :

- Patient has good immunity.
 - Organism has low virulence.
- } Infection contained.

Types :

	Brodie's abscess	Garre's Sclerosing Osteomyelitis
Course	Subacute	Chronic
m/c site	Tibia (Proximal end)	mandible > tibia
Pus	+	-
Clinical feature	• Acute dull aching pain • Low grade fever	Bony swelling
Elevated ESR & CRP	+	-
X-Ray	 Thick sclerotic rim Central lucency	 Thick dense bone (d/t excessive periosteal reaction)
MRi	Penumbra sign : Central lucency surrounded by dense sclerotic rim 	-
Rx	Antibiotics ± Debridement	Antibiotics

Septic Arthritis

00:48:55

Introduction

- Surgical emergency : If pus not removed from joint immediately → Destruction of joint.
- Children (<5 years) > Elderly people (relatively low immunity)

Route :

m/c : Hematogenous

Site :

- m/c site : knee (ends of bone exposed).
- Infants : Hip (Tom Smith arthritis).

Organism :

- m/c : Staphylococcus aureus.
- IV drug abusers: Pseudomonas.
- Sexually active : Gonococcus.

Clinical features :

Young child presents with :

- High grade fever.
- Local inflammatory features :

- Swelling.
- Redness.
- Pain (Effusion in joint → Capsule stretches)

No movement at joint

- Joint in position of ease → ↑ volume of joint.
↓ capsular stretching
↓ pain.

- Signs : Patellar tap sign +ve

Hip : FABER

- Flexion
- Abduction
- External rotation

Knee : 5-30° flexion



Position of ease

Investigations :

1. Blood investigations :

- CBC, ESR, CRP, TLC, blood culture.

2. Radiology

→ X-ray

- Early : No changes ± widening of joint space.
- Late : Destruction of joint.

→ MRI : Synovial thickening.

→ USG guided arthrocentesis : Diagnostic & therapeutic.



Bony ankylosis

Treatment:

Arthrotomy (TOC): Irrigation & debridement of septic joint contents under antibiotic cover.

Complication:

1. Bony ankylosis: End sequelae of neglected septic arthritis.

organisms release proteolytic enzymes: Collagenases, metalloproteinases, elastases.

Destroy articular cartilage → Raw ends of bone exposed → Heal with fusion → Bony ankylosis.

a. Fusion of bone:

Ankylosis (Fusion d/t disease process)

Arthrodesis (Surgical fusion)

Eg: Neglected CTEV → Triple arthrodesis.

	Bony	Fibrous
Fusion between using:	Bony	Fibrous tissue
Causes:	Pyogenic septic arthritis TB spine (Spondylitis)	TB arthritis
movement ↓ Pain:	(-)	(+)
Stability of joint:	Stable	Unstable



Arthrodesis

01:02:59

Tom Smith Arthritis

Septic arthritis of hip in infancy (<1 year).

Emergency: Need immediate intervention to preserve hip functions.

Pathogenesis:

Disease process (Pus & proteolytic enzymes)

Destroy capital femoral epiphysis (Cartilaginous in children)

Chondrolysis.

Clinical features:

- Hyper mobility of hip joint.
- Limb shortening.
- Telescopy test (+).

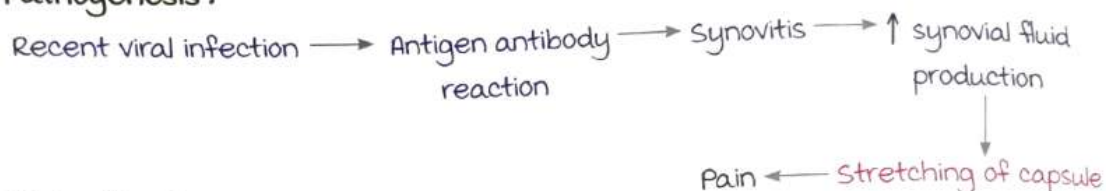


Septic arthritis

----- Active space -----

Transient Synovitis

- Self limiting inflammation of synovium.
- Age group : 6-12 years.

Pathogenesis :**Clinical features :**

- Pain.
- Limp.
- mild fever.
- Position of ease.

management :**Investigation :**

- Blood investigation : ESR, CRP, TLC.
- Aspiration of joint : **Best investigation** to differentiate from septic arthritis.

Treatment : Observation (Transient synovitis also called observation hip).

Kocher's criteria :

To differentiate between transient synovitis & septic arthritis.

- Difficulty in weight bearing/unable to walk.
- Temperature $>38.5^{\circ}\text{C}$ (101.3°F).
- ESR >40 mm/hr
- WBC $>12,000$ cells/ mm^3 .

- Each criteria has 1 point
- ↑ point → ↑ chance of septic arthritis.

Note :

Cause of limp in child : Transient synovitis (m/c) > septic arthritis > Perthes' disease

Infections of Hand**Paronychia (Running around the nail) :**

- m/c infection of hand.
- Infection around the nail and nail bed.

m/c organism : Staphylococcus aureus.



Paronychia

Treatment :

Antibiotics + drainage of abscess + removal of part of nail (if infection reaches nail bed).

----- Active space -----

Felon/whitlow :

Infection of subcutaneous tissue (Pulp) of the finger (Distal phalanx).

Finger affected : Thumb (m/c) > index finger.

m/c organism : Staphylococcus aureus.

Complications :

- Osteomyelitis of phalanx.
- Tenosynovitis.

Treatment : Antibiotics + incision (vertical incision to break adhesions/septae that hide infection) & drainage.



Felon/whitlow

Tenosynovitis :

Infection of tendon/tendon sheath.

m/c organism : Staphylococcus aureus.

Clinical features :

- Symptoms :
 - Fever.
 - Flexion attitude.
 - Fusiform swelling of fingers.
 - Pain on passive extension.
- Signs : Kanavel sign +ve (Pain on tapping affected tendon).



Tenosynovitis

Treatment : Antibiotics + incision & drainage



Tenosynovitis

----- Active space -----

TUBERCULOSIS OF BONE AND JOINTS

INTRODUCTION

AKA *Caries/koch's disease*.

Less severe local manifestation compared to pyogenic bacteria.

Faucibacillary infection by Koch's weak bacillus.

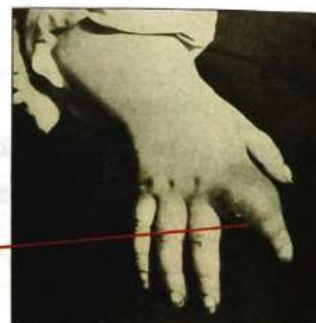
Organism : *Mycobacterium tuberculosis*. Route : Hematogenous (From 1° source).

m/c site :

- In the body : *Lung*.
- In musculoskeletal system : *Spine* (TB spondylitis) > hip > knee.
- In spine : *Dorsolumbar* > dorsal > lumbar
 - In children : Cervical spine.

Variants :

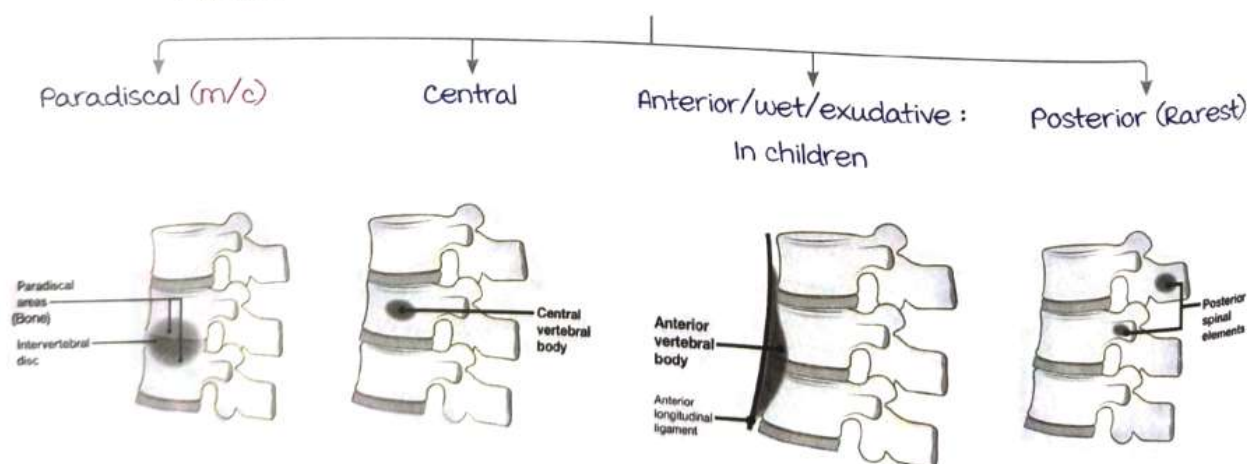
- *Caries sicca* : TB shoulder.
 - Dry/non-exudative TB
- *Spina ventosa* : TB fingers.
 - 'Filling up of air' like appearance of fingers.



TB Spine/TB Spondylitis/Pott's Spine

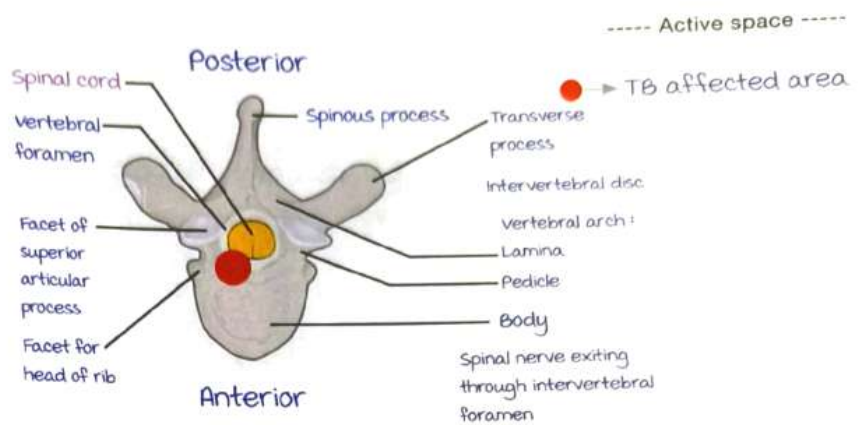
00:02:53

TYPES :



Paradiscal :

- 2 vertebrae involved :
Lower half of upper vertebrae + disc + upper half of lower vertebrae.
(D/t development from common somite/sclerotome → Common blood supply).
- Anterior to spinal cord.



CLINICAL FEATURES

Constitutional features :

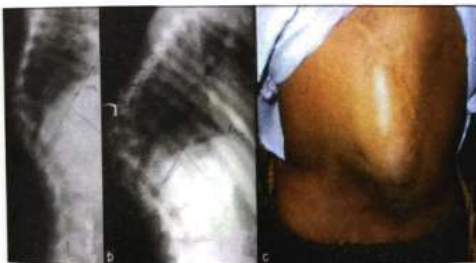
- Low grade fever.
- Malaise.
- Night pains.
- Evening rise of temperature.

Local findings :

- Back pain (Earliest symptom) & Paraspinal spasm/tenderness (Earliest sign)

military attitude ← Cautious gait

- Cold abscess.
- Deformity (D/t vertebral collapse)
Spinous process felt prominently on palpation :
- Knuckle : Prominence of 1 spinous process.
- Gibbus : Prominence of 2-3 spinous process.
- Angular Kyphosis : Prominence of >3 spinous process.

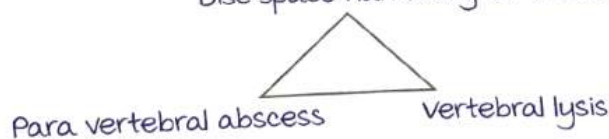


Gibbus formation

INVESTIGATION & COMPLICATIONS :

1. X-ray :

Disc space narrowing (D/t destruction/dessication of disc) : Earliest



MRI

2. MRI : IOC

- Disc involved.
- Paradiscal disease involving bone and cartilage.

3. CT guided biopsy : Gold standard
Organism identified from tissue.

Complication/end sequelae :

Neglected TB spine → Bony ankylosis.

Note : TB anywhere else → Fibrous ankylosis.

----- Active space -----

POTT'S PARAPLEGIA

TB spine (Pott's spine) → Compresses on spinal cord → Neurological manifestations.

Causes :

1. Mechanical : D/t granulation tissue.
2. Cord edema : D/t inflammation
3. Thrombosis of spinal artery → Ischaemia of spinal cord → Rapid onset paraplegia (Poor prognosis).



MRI : Cord compression

Clinical features :

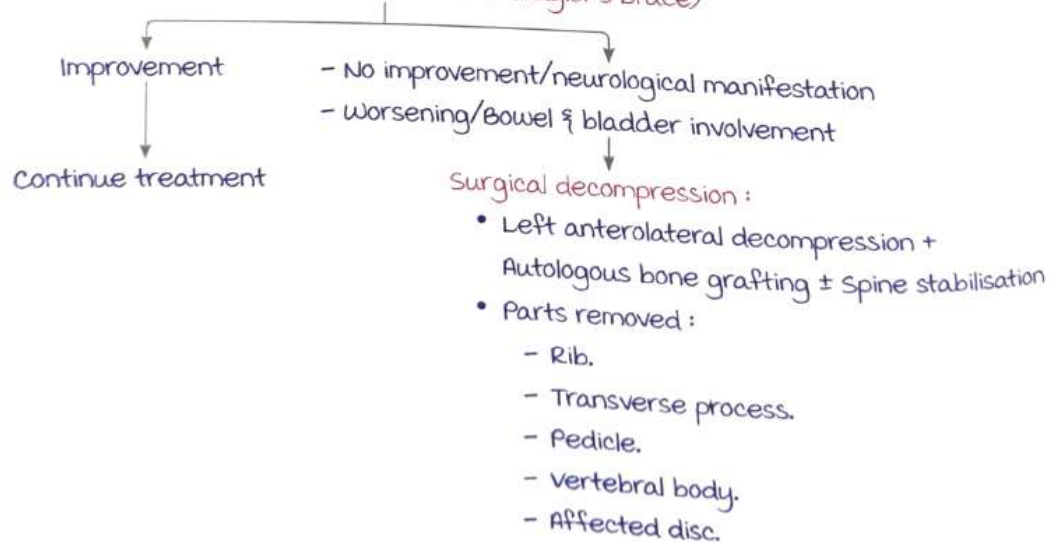
- Early neurological manifestations (UMN finding)
 - Ankle clonus/exaggerated deep tendon reflexes.
 - Extensor plantar.
- Late : Bowel & bladder involvement → Poor prognosis.

Prognostic factors :

	Better prognosis		Bad prognosis	
Age	Younger		Older	
Speed of onset	Slow		Rapid	
Onset	Early	Compression d/t granulation tissue	Late	Compression d/t bony deformity
Duration of cord involvement	Shorter history		Longer (>12 months)	
Vertebral disease	Active/exudative/wet		Healed	

TREATMENT**middle path regimen :**

ATT for 18-24 months + rest + brace (Taylor's brace)



TB SPINE VS METASTASIS (D/D)

----- Active space -----

MRI :

Disc involved (Bad disc) → TB spine (Good prognosis).
 Disc spared (Good disc) → metastasis (Bad prognosis).



MRI : TB spine (Disc involved)



MRI : metastasis (Disc spared)

X-ray :

metastasis : **winking owl sign** (D/t destruction of pedicle)



Absent pedicle →



Winking owl sign

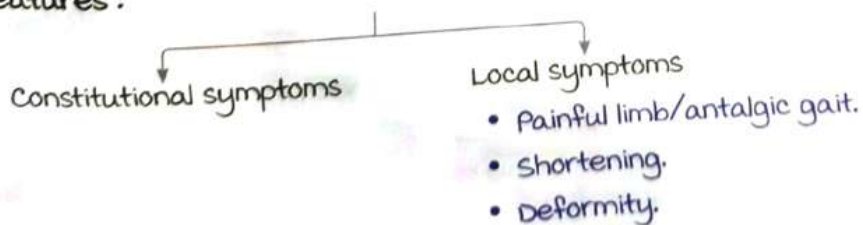
00:21:16

TB Hip

Site/first focus of appearance :

- Acetabulum.
- **Babcock's triangle**.
 - Area in neck of femur.
 - Deficient in monocyte & macrophage → Organism grows easily.

Clinical features :



----- Active space ----- **Stages :**

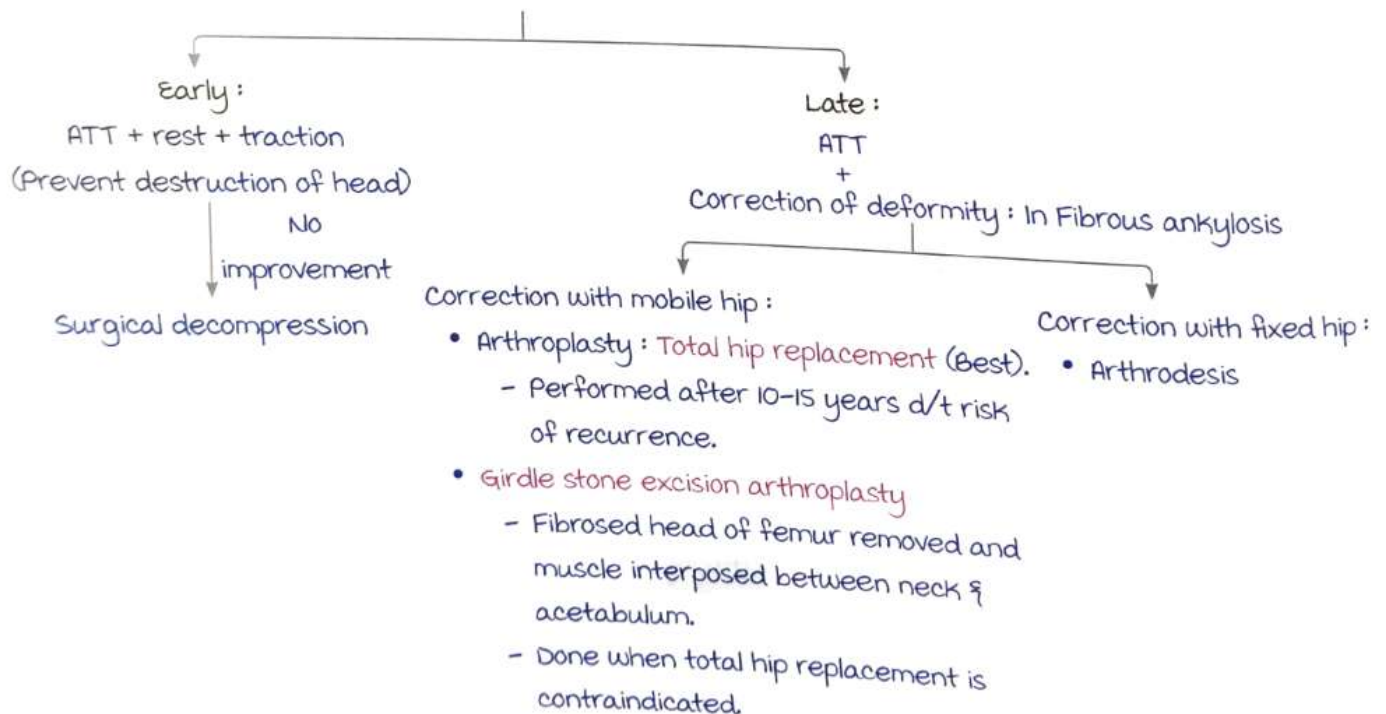
Stage I	Synovitis/stage of apparent lengthening	Inflamed synovium $\rightarrow$ $\uparrow$ synovial fluid secretion $\rightarrow$ distension of capsule $\rightarrow$ pain $\rightarrow$ position of ease (Flexion, abduction, external rotation)
Stage II	Early arthritis/stage of apparent shortening	- Destruction of articular cartilage begins - Flexion, adduction, internal rotation - Apparent shortening < 1 cm
Stage III	Late arthritis/stage of true shortening	- Significant destruction of articular cartilage - Flexion, adduction, internal rotation - True shortening > 1 cm
Stage IV	Advanced arthritis/wandering acetabulum	Destruction of head of femur (Small head against large acetabulum) $\rightarrow$ Pestle and mortar deformity
Stage V	Fibrous ankylosis	Painful, unstable hip with limited mobility

Investigation :

X-ray :

Phemister triad :

1. Juxta articular osteopenia/osteoporosis : **Earliest** sign in X-ray
2. Peripheral osseous erosions.
3. Narrowing of joint space.

Treatment :

TB knee

00:27:48

----- Active space -----

Clinical features :

Constitutional symptoms

Local symptoms :

- Painful limb/Analgic gait.
- Attitude of knee : $5-30^\circ$ flexion (Postion of ease d/t initial synovitis leading to effusion).
- Shortening.
- Deformity :

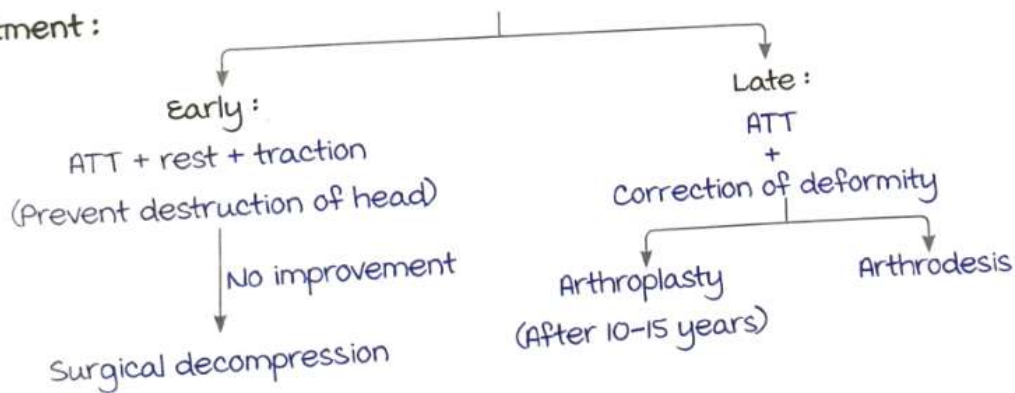
Triple deformity in late stage : PERF

1. Posterior subluxation of knee.
2. External Rotation of tibia.
3. Flexion of knee.

Investigations :

- MRI (IOC) : Synovitis seen.
- Biopsy : Gold standard.

Treatment :



Note :

- Non healing sinus in chronic osteomyelitis : D/t presence of sequestrum.
- Periosteal reaction not seen in tubercular osteomyelitis.
- Part of vertebrae least involved in TB spine : Spinous process.
- Paraplegia in TB spine commonest in upper thoracic region (Spinal cord thicker → Easier to compress).
- Charcot's joint seen in :
 - Syphilis.
 - DM.
 - Syringomyelia.

----- Active space -----

PAEDIATRIC ORTHOPAEDICS : TRAUMA, NECK AND UPPER LIMB

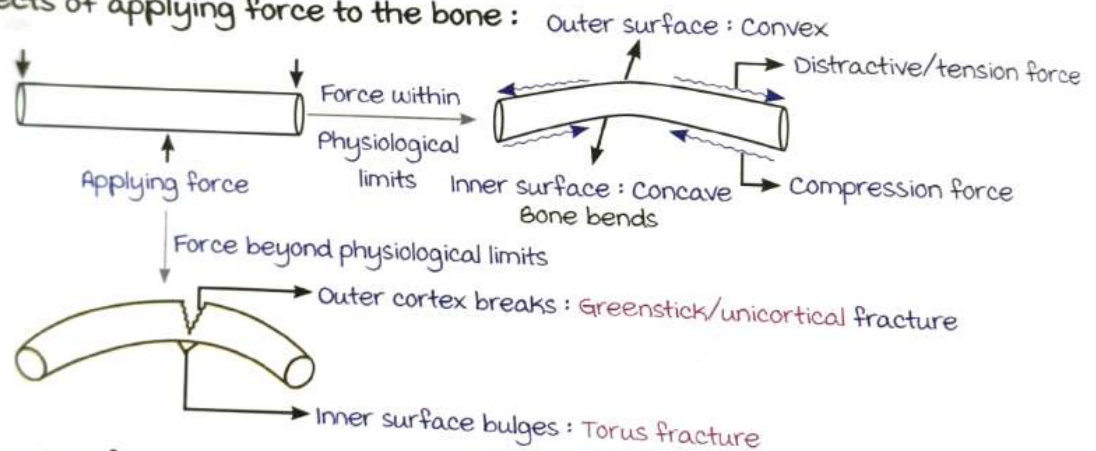
Characteristics of paediatric bones :

1. more water content :
 - Flexible and resilient to stress.
 - Bends > breaks.
2. Thicker periosteum.
3. Potential to remodel & correct deformities and fractures (Especially metaphysis).

Paediatric Fractures

00:02:52

Effects of applying force to the bone :



Greenstick fracture :

- m/c fracture occurring in children.
- mimics stem of a growing tree.
- Location : Radius > ulna.

Treatment : Break the other intact cortex (Osteoclasis) → Realign the bone.



Greenstick fracture



Torus fracture

Torus fracture :

- Occurs d/t buckling of bone.
- Resembles a tubular cylinder at base of a pillar.

Treatment : POP application.

Note :

1. m/c fracture at birth : Clavicle.
2. m/c fracture in a child after fall on outstretched hand : Supracondylar fracture.
3. m/c fracture overall : Clavicle.

Toddler's fracture

Spiral fracture of tibia.

Etiology : Trivial accidents.

Features :

- Intact fracture pattern (Undisplaced).
- Well supported fracture (D/t thick periostium).
- Fibula is intact so child can present walking.
 - Can bear weight on fractured bone.
 - Limp d/t pain.

Treatment : POP cast.



X-Ray : Spiral fracture

----- Active space -----

Battered Baby Syndrome

00:10:52

Features :

- multiple fractures (usually metaphyseal) or bruises in different stages of healing.
- Inconsistent history, signs of neglect.
- Long bone fractures in non-ambulatory children.
- Suspicion of battered baby syndrome → X-ray Scanogram.

metaphyseal bucket handle fracture

D/t violent to and fro shaking of children < 2 yrs age

→ Avulsion of capsule from the bone.



metaphyseal bucket handle fracture

Growth Plate Injuries

00:12:43

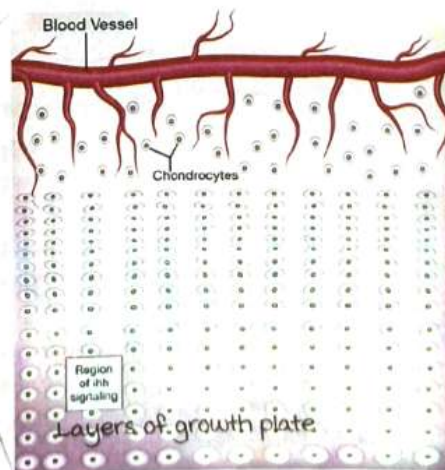
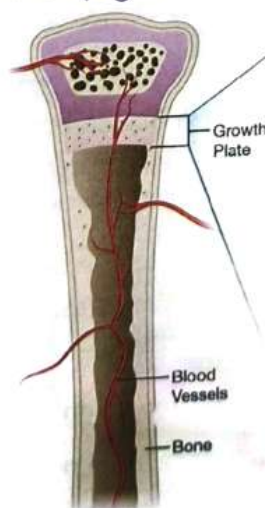
Direction of growth : Epiphysis to metaphysis.

Layers of growth plate :

1. Germinal : most important.
2. Proliferative.
3. Hypertrophic : weakest.
4. Calcification.
5. Ossification.



Growth plate



Growth plate zones

Chondrocyte stem cells in germinal zone

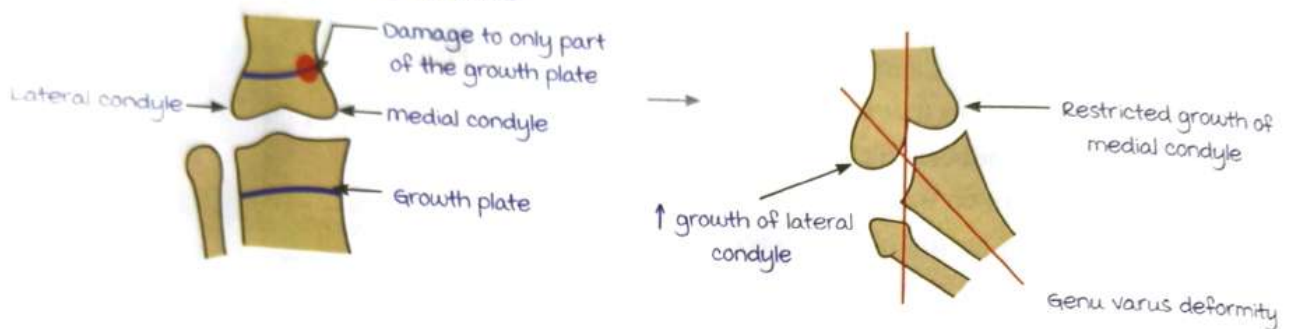
Proliferation zone

Zone of maturity and increase in cell size

..... Active space

Injuries affecting **germinal layer** → Impairs growth → Limb length discrepancy, Deformity.

Example : At the knee



Salter & Harris classification of physeal/growth plate injuries :

Type	Image	Features	Prognosis	Examples
I : Split fracture		- Fracture splits the growth plate without injuring germinal layer. - Normal growth on reducing the fracture.	Good	Slipped capital femoral epiphysis (SCFE)
II : m/c		- Fracture splits the growth plate. - Fracture line goes towards the metaphysis. - No injury to the germinal layer. - Normal growth on reducing the fracture. - metaphyseal fragment : Thurston holland.	Good	Supracondylar humerus fracture
III		- Fracture line splits the growth plate and goes towards the epiphysis. - Germinal layer will be injured. - No growth on reducing the fracture.	Bad	Fracture of lower end of tibia
IV		- Fracture line through all layers. - Germinal layer injured. - Growth is impacted even on reducing the fracture.	Bad	Lateral condyle of humerus fracture
V		- Impaction injury → Crushes growth plate. - Fracture is missed on X ray. - Retrospective diagnosis. - Late presentation with limb length discrepancy.	Worst	Fall from height

Paediatric Deformities

00:25:02

----- Active space -----

CONGENITAL MUSCULAR TORTICOLLIS/WRY NECK

- Tilting of head to one side.
- most commonly occurs on **right side**.

Etiology :

Birth trauma of sternocleidomastoid muscle.

- Breech presentation.
- Shoulder dystocia.
- Forceps delivery.

Pathogenesis :

- Contraction of sternocleidomastoid → Neck tilt.
- Ischemia of SCM → Necrosis of fibrosis → Contracture
 - At junction of upper 2/3rd and lower 1/3rd.
 - Affects **clavicular head** of SCM.

Clinical features :

- Only head tilt is appreciated soon after birth.
- Fibrotic mass maybe palpated 3 months after birth (SCM tumor).

Treatment :

1. Conservative :
 - Gradual stretching of the neck → Corrects itself.
 - most cases are self resolving.
2. Surgical release : If not corrected by 1 year of age.

Complication :

If not corrected by 1 year of age → **Plagiocephaly** (Head grows in an abnormal direction).

Note :

Acquired causes of torticollis : vision problems, extra ocular muscle palsy

SPRENGEL'S DEFORMITY

- Congenital undescended hypoplastic scapula.
- Associated with **Klippel Feil syndrome**.

Etiology :

Presence of **omovertebral bar** → Prevents scapular descent.



Congenital torticollis



Cock robin appearance



Sprengel's deformity : undescended scapula

----- Active space -----

Features :

- One scapula at a higher level than the other.
- Restricted movement of affected shoulder.

Treatment :

- Woodward operation (Between 3-8 years of age).
- If done after 8 years of age → Neural traction injury.

Klippel Feil syndrome :

Strong association with scoliosis.

Cause : **Segmentation defect** of cervical vertebrae.

Features :

- Short webbed neck.
- Restriction of neck movements.
- Low set hair line.

MADÉLUNG DEFORMITY

- Ulna radial mismatch.
- 50% cases are bilateral.

Etiopathogenesis :

Defect in the growth plate of distal end of radius.

- Affects radius growth.
 - Normal ulna growth.
- } **Ulna radial mismatch**



made lung deformity ulnar radial mismatch

Clinical features :

- Prominent ulna dorsally.
- **Dinner fork deformity** in children.

Treatment :

Darrach's procedure : Resect terminal end of ulna.

Limb Deficiency**TYPES****1. Amelia :**

- Both long and short bones absent/complete absence of limb.

2. Phocomelia :

- Long bones absent, short bones present.
- Cause : Thalidomide.



Amelia

00:32:38

3. Hemi-melia :

- Longitudinal deficiency of limb.
- m/c : **Fibular hemimelia**.
- Tibial hemimelia : Tibia is absent.
- Radial hemimelia/radial club hand : Absent radius.
 - Longitudinal deficiency of radial side of limb.

----- Active space -----



Fibular hemimelia

Absent fibula



Radial hemi-melia

Absent radius

Associated syndromes :

1. Holt-Oram syndrome.
2. Fanconi syndrome.
3. Tar syndrome.
4. VACTER syndrome.

MISCELLANEOUS

Cleido cranial dysostosis :

- Abnormality in **skull and clavicle**.
- Congenital absence of clavicle.
- Shoulders can be brought close to each other.



Absence of clavicle



Cleidocranial dysostosis

Poland syndrome :

- Congenital absence of muscle.
- m/c : **Pectoralis muscle absence**.



Poland syndrome

----- Active space -----

PAEDIATRIC ORTHOPAEDICS : HIP

Paediatric hip :

Features :

1. Shenton's line :

- Inferior margin of superior pubic ramus (Laterally) → medial part of head and neck.
- Pathology causes disturbance in Shenton's line.

Shenton's line

Neck shaft angle



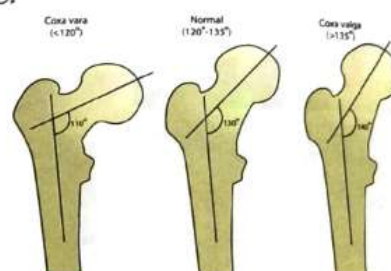
xray of hip

2. Neck-shaft angle :

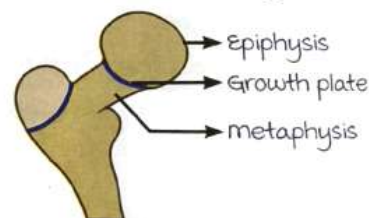
- Normal is $120-130^\circ$ (Adults).
- ↑ neck shaft angle : Coxa valga.
- ↓ neck shaft angle : Coxa vara.

3. Proximal femur in children has 2 growth plates.

- Capital femoral epiphysis : Head of the femur epiphysis.



Neck shaft angle



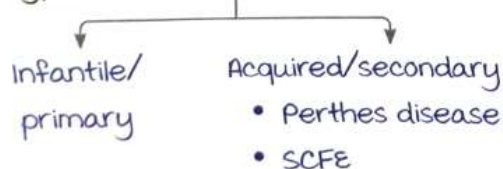
2 growth plates

Coxa Vara

00:01:20

Injury to the growth plate of head of femur → Greater trochanter continues to grow → Head and neck stop growing → Coxa vara

Types of coxa vara :



Clinical features :

- Trendelenburg gait : Failure of hip abduction.
- Obligate external rotation : External rotation of hip (knee goes out) on flexion of hip.
- Limitation of abduction and internal rotation.
- Limb shortening.



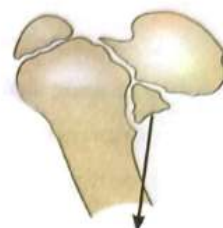
Obligate external rotation

CONGENITAL COXA VARA

Etiology :

- Damage to fairbank triangle.
- Fairbank triangle : Part of neck of femur.

Note : Babcock's triangle is seen in TB.



Fair bank triangle

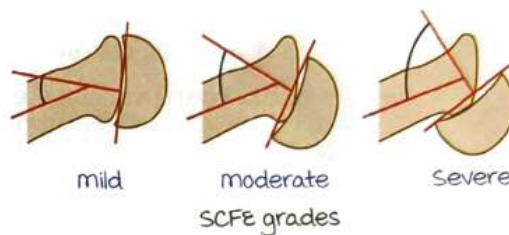
SCFE (SLIPPED CAPITAL FEMORAL EPIPHYSIS)

misnomer : Head remains inside the acetabulum whereas neck slips.

Pathogenesis :

Fracture of the growth plate (Type I Salter-Harris)

Head and neck are disrupted



mild

moderate
SCFE grades

Severe

Head remains in the acetabulum and neck slips.

Etiology :

- Idiopathic : m/c cause.
- Associated with endocrinopathy (D/t sex hormone : GH imbalance in puberty).
 - m/c : Hypothyroidism, hypogonadism, $\uparrow$ GH.

Bone growth

Bone fusion

General features :

- male > female.
- Risk factor : more weight for age in children.
- 40% cases are bilateral (60% cases if associated with endocrinopathy).

Clinical features :

Presents at puberty

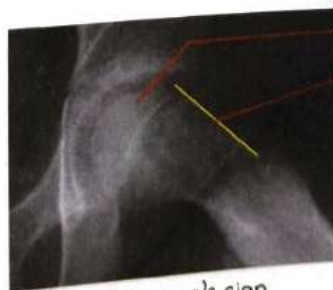
Males : 13-17 yrs

Females : 11-14 yrs

- Hypogonadism
 - Voice change
 - Gynecomastia
 - Coxa vara features.
- In endocrinopathy features

Trethowan's sign on X-ray :

Line drawn on the lateral aspect of neck of femur does not intersect with head of femur.



Trethowan's sign

Epiphysis

Kein's line : Does not intersect with head of femur

Treatment :

Closed reduction + internal fixation + in situ pinning.

----- Active space -----

Osteochondritis/Osteonecrosis

00:15:31

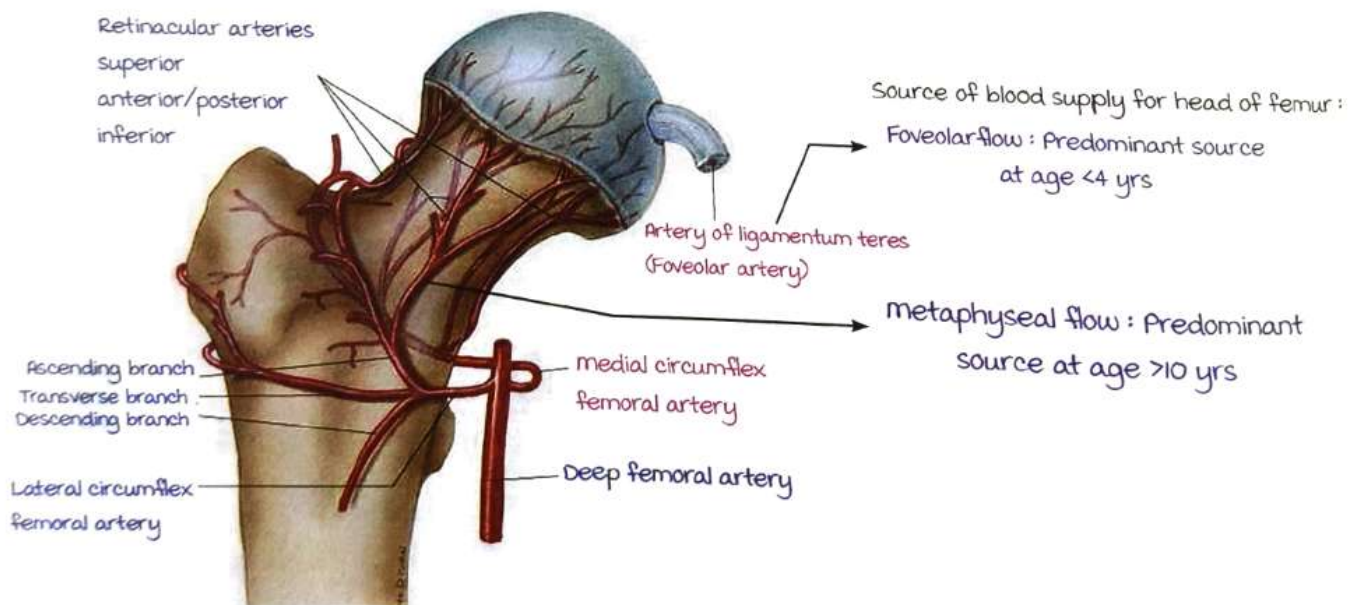
- Osteochondritis is a self-limiting abnormality of bone growth.
- Involves centres of ossification in the epiphysis.
- Begins in childhood as a **necrotic** condition.
- D/t primary **vascular** event or **repetitive trauma**.

Osteochondritis	Bone
Perthes disease (m/c)	Femoral epiphysis
Kohler's disease	Navicular bone
Keinbock's disease	Lunate
Panner's disease	Capitulum (Elbow pain)
Osgood-Schlatter's disease	Tibial tuberosity
Sever's disease	Calcaneal epiphysis

PERTHES DISEASE

AKA coxa plana/Legg-calve-perthes disease.

- Idiopathic, spontaneous osteonecrosis (Chandler's disease in adults).

Blood supply of femur :

Bridge phase between 4-10 yrs (Shift from foveolar to metaphyseal)

Head of femur is vulnerable

Any insult

Osteonecrosis

General features :

- m/c cause : Idiopathic.
- m/c association : Protein S deficiency.
- male > female
- 10-12% cases : Bilateral.
- Age > 6 years : Poor prognosis.

Clinical features :

- Painless limp, can be painful later.
- Pain radiates to knee/groin.
- Limitation of abduction and internal rotation.
- Obligate external rotators.

Investigations :

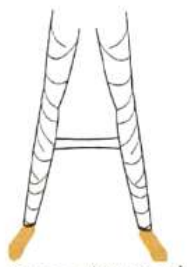
- MRI : IOC.
- X-Ray : Late change.



Osteochondritis of navicular bone



Oschgood schlatter



Broomstick cast



Osteochondritis of lunate

Treatment :

- Sustain the head till vascularity resumes.
- No weight bearing.
- Strict skin traction.
- Petrie cast/broom stick cast.
- Total hip replacement after skeletal maturity.

Development Dysplasia of Hip

00:24:58

- Previously called congenital dislocation of hip.
- Idiopathic spontaneous subluxation of femoral head.

Pathogenesis :

Flat/convex/shallow acetabulum



Head of femur lies outside the acetabulum (In utero).

Epidemiology :

- 1:1000 live births.
- Females > males.
- 20% cases are bilateral (B/L).
- Left side > B/L > right side.
- Usually seen in 1st born female child with fetal malformation (m/c breech).
- Strong family history.
- Caucasians > asians → D/t the way babies are carried in infancy.

----- Active space -----

Clinical features :

- Asymmetry of limbs : Shortening of affected side.
- ↑ thigh and gluteal folds.

Vascular sign of NARATH :

- Femoral pulses are not felt (As head of femur does not lie against it).
- Popliteal pulse can be felt.

Clinical diagnosis :

Performed in child <3 months age.

1. Barlow's test :

- mnemonic : **BAD**.
- Adduction : Hip is dislocated.

2. Ortolani's test :

- Abduction : Dislocated hip is reduced.

3. Allis/Galeazzi sign :

- Flexion of both hip and knee.
- Dislocated side has knee at a lower level.
- Only seen in U/L DDH.

4. Klisic test : For B/L DDH.

5. Trendelenburg gait/waddling gait



Ortolani's test

Investigations :

- Screening : USG for <6 months age.
- IOC : **MRI** (To confirm diagnosis & plan management).

Acetabulum in children :

- Triradiate cartilage (made of ilium, ischium, and pubis).
- **Hilgenreiner's line** :
 - Line passing through centers of both triradiate cartilage.
- **Perkins line** :
 - Line perpendicular to the Hilgenreiner's line.
 - Passes through lateral edge of acetabulum.
 - Head of femur
 - Normal : Inner, outer quadrant.
 - DDH : upper, outer quadrant.



Hilgenreiner's and Perkins line



von russen splint

Treatment :

Age	Treatment
0-6 months	Pavlik harness, von rosen splint
6-18 months	Reduction : Open/closed Smith peterson approach
18-36 months	Femoral osteotomy
>3 years	VDRO + pelvic osteotomy (Salter, Pemberton)
>10 years	Total hip replacement after skeletal maturity

----- Active space -----

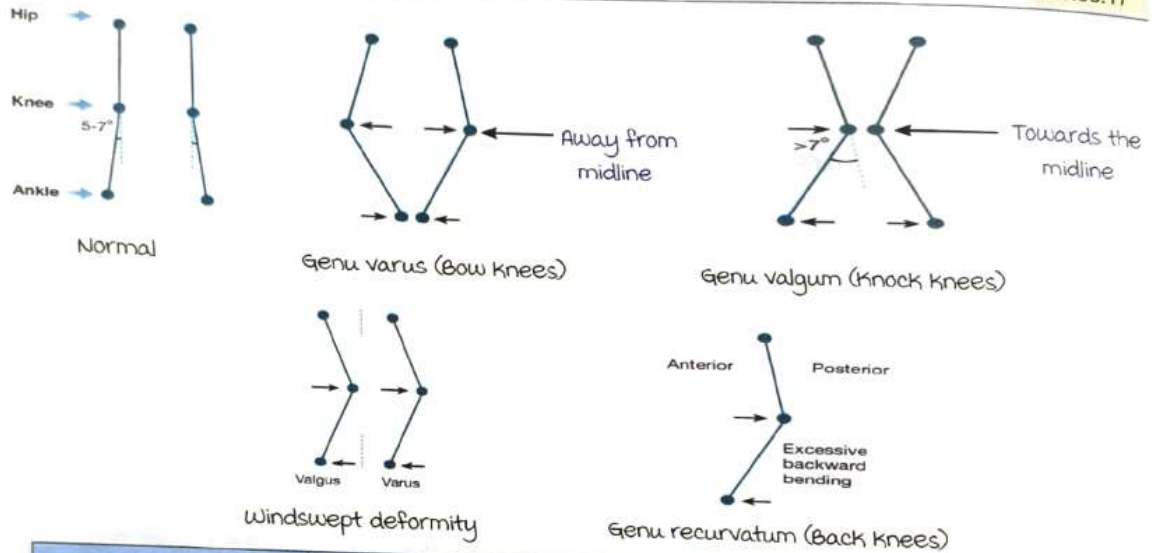


Pavlik harness

PAEDIATRIC ORTHOPAEDICS : KNEE AND FOOT

Deformities of Knee

00:00:17



	B/L genu varum (Bow legs)	B/L genu valgus	Wind swept deformity
m/c cause in children	Rickets > Idiopathic	Idiopathic > Rickets	Rickets
m/c cause in adults	Osteoarthritis > Rheumatoid arthritis	Rheumatoid arthritis > Osteoarthritis	Rheumatoid arthritis



Genu varum



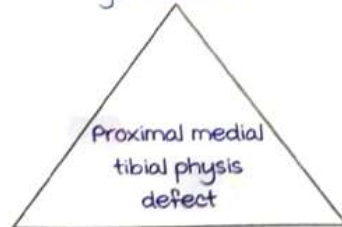
Genu valgus



Windswept deformity

Blount's disease/infantile tibia vara :

B/L idiopathic genu varum



Sign : Siffert Katz sign

Internal rotation of tibia

X-ray :

metaphyseal beaking



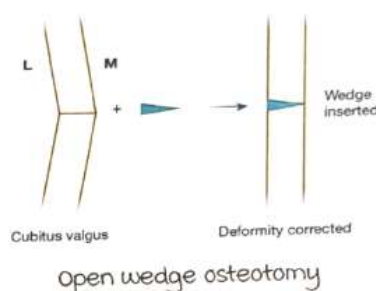
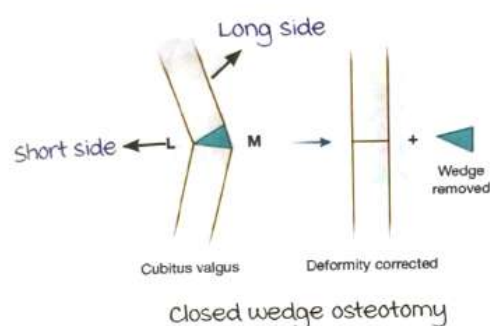
Blount's disease

causes of deformity :

- malunion.
- Growth plate injury.

Treatment :

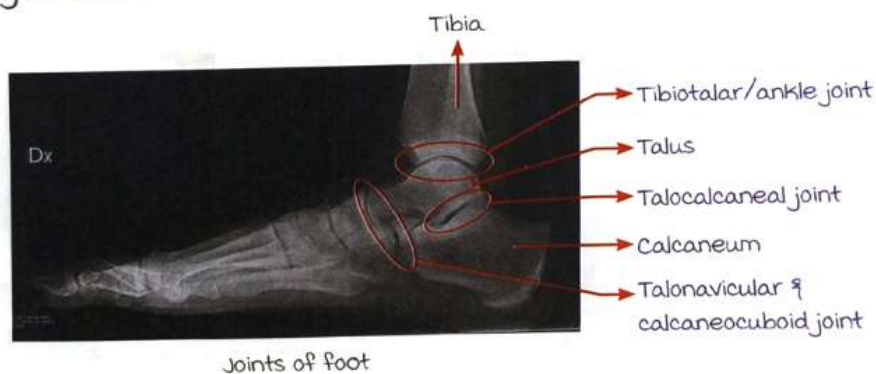
- open wedge osteotomy : Lengthen the short side by filling extra bone.
- closed wedge osteotomy : Shorten the long side by removing bone.



Deformities of Foot

00:09:35

Anatomy of foot :



Actions of the joints :

- Tibio-talar/ankle joint : Ankle plantar/dorsiflexion.
- Talocalcaneal/subtalar joint : Inversion/eversion of foot.
- Talonavicular & calcaneocuboid/midtarsal joint : Forefoot adduction/abduction.

Note :

medial longitudinal arch of foot : Provides concavity of the foot

Lost : Flat foot.
Exaggerated : Cavus of foot.

----- Active space -----

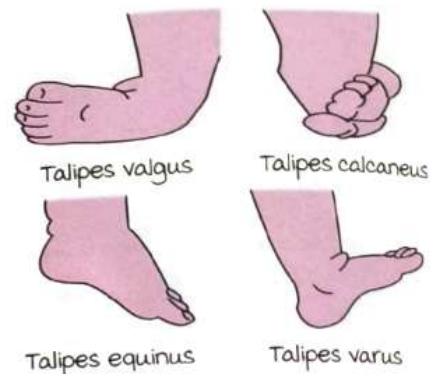
Deformities of the joint :

Joint	Position of deformity	Term of deformity
Ankle joint/Tibio talar joint	Ankle plantar flexion	Equinus deformity
	Ankle dorsiflexion	Calcaneus deformity
Subtalar/talocalcaneal joint	Inversion of foot	varus deformity
	Eversion of foot	valgus deformity

Terminology :

Talipes : Congenital deformity in which foot is twisted.

- Talipes cavus : High arched foot.
- Talipes equinus : Plantar flexion.
- Talipes calcaneus : Foot dorsiflexion.
- Talipes valgus : Abduction and eversion.
- Talipes varus : Adduction and inversion.

**Congenital Talipes Equinus Varus (CTEV)**

00:14:30

AKA **club foot** (Appears like a golf club).
m/c congenital abnormality of leg, ankle & foot.

Epidemiology :

- 50% : Bilateral.
- 1:1000 live births.
- males > females.
- 1st born child a/w fetal malpresentation like **breach**.



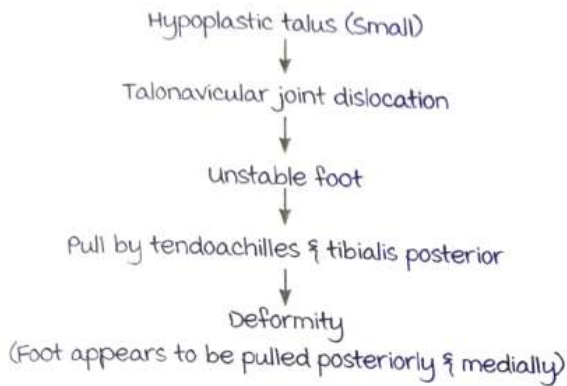
CTEV

Etiology :**1. Congenital/primary CTEV : m/c type**

- m/c cause : **Idiopathic**.
- Spina bifida.
- Arthrogryposis multiplex congenita (muscle fibrosis → contracture → deformity).

2. Acquired causes : Neuromuscular disorders (Poliomyelitis).

pathoanatomy :

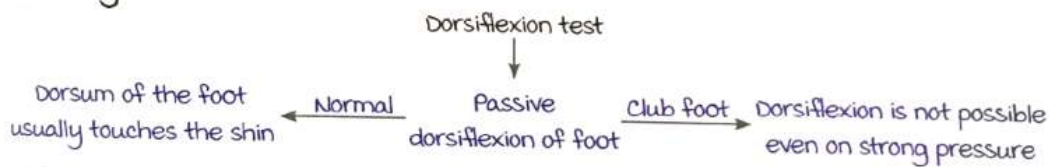


----- Active space -----

Primary deformities of the foot :

- C**
 ↓
Cavus
 Exaggeration of medial longitudinal arch
 - A**
 ↓
Adduction
 At talonavicular/ mid tarsal joint
 - V**
 ↓
Varus
 At talocalcaneal/ subtalar joint
 - E**
 ↓
Equinus
 At ankle joint
- Internal rotation of tibia is also seen.

Screening test :

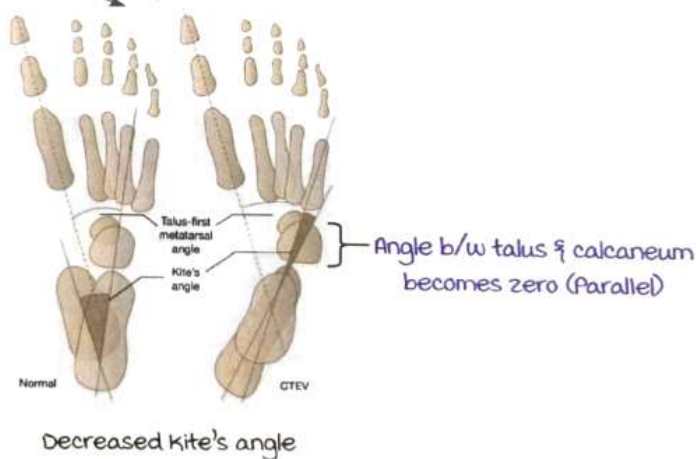


Confirmatory test :

Kite's angle : Talocalcaneal angle



Normal Kite's angle



----- Active space -----

Treatment in newborn :

- Start treatment by 2 weeks.
- manipulation of foot to correct deformity → maintain with POP (7-8 weeks).

Order of manipulation : Ponseti method
C → V & A → E

- Above the knee.
- Keep for 1 week.
- Repeat procedure after 1 week.

	Ponseti's method	Kite's method
Treatment starts	<2 weeks	1 month
Order of correction	Cavus → adduction → varus → equinus	Adduction → inversion → equinus
Fulcrum	Head of talus	Calcaneocuboid joint
POP type	Above knee cast	Below knee cast
Duration	6-8 weeks	6-9 months

Rocker bottom foot :**Causes :**

- Overcorrection of CTEV.
- Congenital vertical talus.



Rocker bottom foot



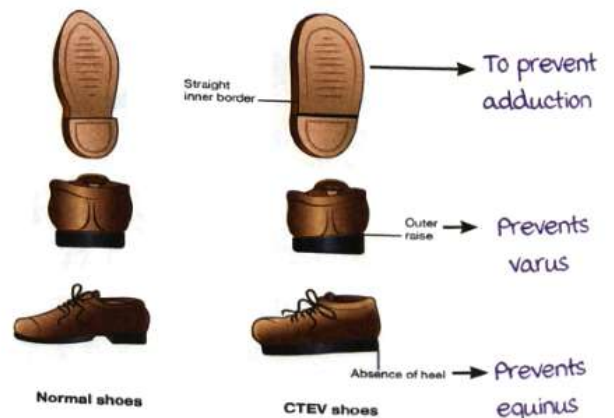
Dennis brown splint

Prevention of recurrence :**1. Dennis brown splint :**

- Till the baby learns to walk.
- mechanism : As baby moves the leg, deformity keeps getting corrected.
- Disadvantage : Low compliance.

2. CTEV shoes :

- Once the baby starts walking (up to the age of 3-5 yrs).
- use Dennis brown splint when baby sleeps.
- Should be worn at least 23 hrs/day.

**Treatment in late presentation :**

1. 1-3 yrs : Posteromedial soft tissue release $\xrightarrow{F/b}$ Ponseti manipulation.

- Soft tissues contract by 1 year of age.
- Turco's procedure : To release the soft tissue (Tendoachilles, tibialis posterior, flexor digitorum longus).

2. 3-5 yrs :

- By 3 yrs forefoot adduction becomes bony.
- PMSTR + Dilwyn Evans' procedure.
- Dilwyn Evans' procedure : Lateral column shortening by Calcaneocuboid fusion.

----- Active space -----

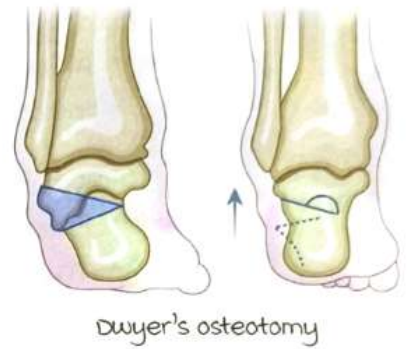
3. 5-8 yrs :

- By 5 years varus becomes bony.
- Dilwyn's Evans' + Dwyer's osteotomy (Wedge osteotomy).

4. 8-10 yrs : Wedge tarsectomy.

5. >10 yrs : Triple arthrodesis

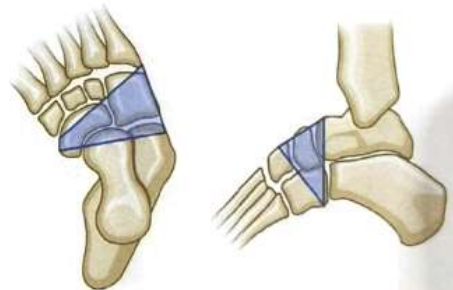
- Surgical fusion of talonavicular, talocalcaneal and calcaneocuboid joint.
- Complication : If performed below 10 yrs of age → Arthrodesis fails
→ Pseudoarthrosis.



Dwyer's osteotomy

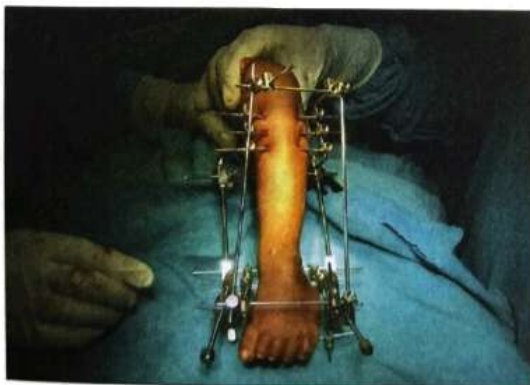


Triple arthrodesis



Wedge tarsectomy

Deformity correction device : Joshi's external stabilization system.
Based on Ilizarov's principle of external fixation.



Joshi's external stabilization system

----- Active space -----

Achondroplasia

00:33:10

m/c cause of **disproportionate dwarfism**.

Etiology :

- **FGFR₃** gene mutation on **chromosome 4**.
- **Autosomal dominant** inheritance or sporadic.

Pathogenesis :

- Affects **enchondral ossification** of **long bones**.
- Intramembranous ossification is normal (Flat bones of skull & clavicle).



Achondroplasia

Clinical features :

- Proximal limb shortening (**rhizomelic shortening**).
- Trunk is usually normal.
- Patients have **normal IQ and sexual development**.
- Brachydactyly : Short, stubby fingers.
- Frontal bossing of skull.
- Starfish hand.
- Trident hand.
- Bullet nose vertebrae.
- Champagne glass pelvis.
- Saddle nose.



Frontal bossing



Brachydactyly



Bullet nose vertebrae

Note :

- Shortening of middle part of limb : **mesomelic**.
- Shortening of distal part of limb : **Acromelic**.

SPINE : GENERAL AND TRAUMA

----- Active space -----

Spine : Components

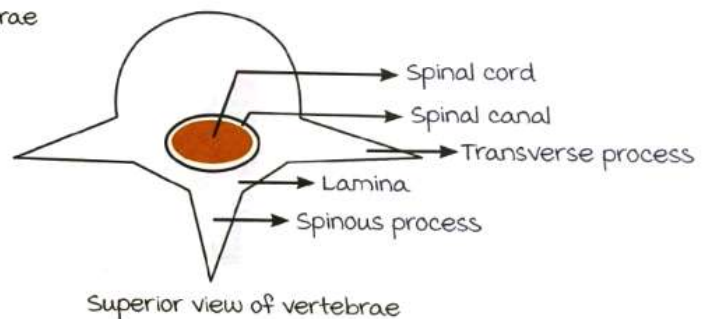
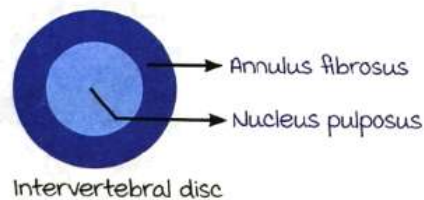
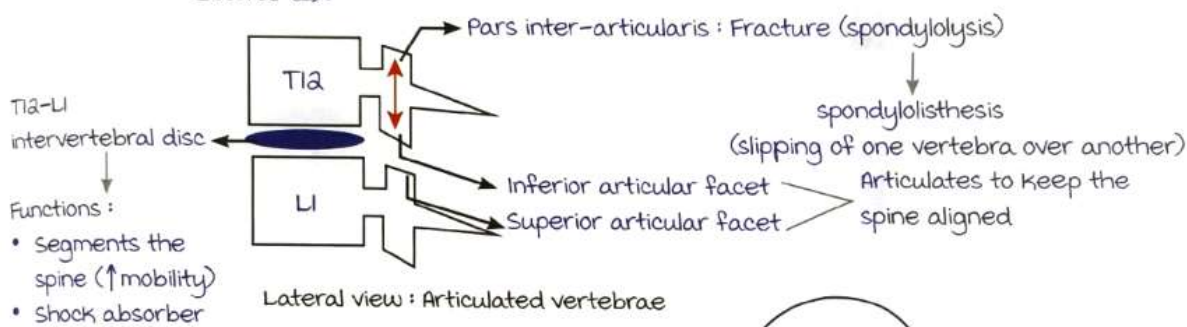
- musculoskeletal : vertebral column
- Neurological : spinal cord + spinal nerves

Anatomy of musculoskeletal elements of Spine

00:01:25

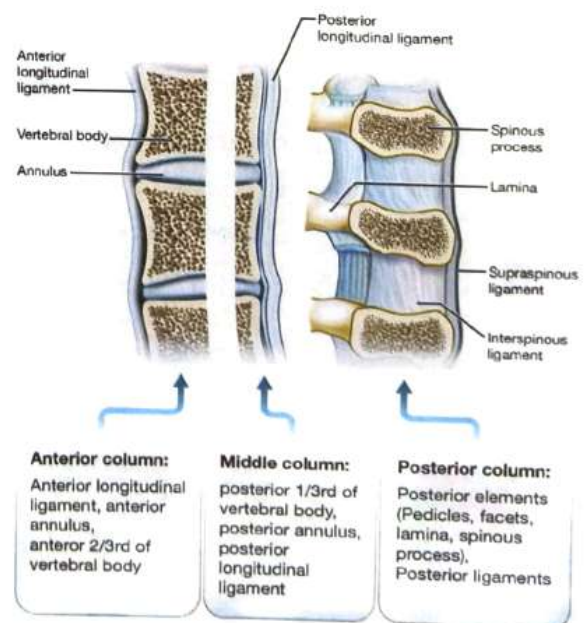
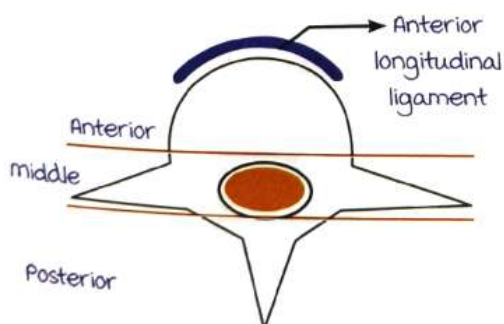
vertebrae

- Cervical (C).
- Thoracic (T)/Dorsal (D).
- Lumbar (L).
- Sacral (S).



DENIS' CONCEPT OF SPINE STABILITY

Based on 3 columns of vertebra :



----- Active space ----- Clinical aspect :

**Trauma to spine**

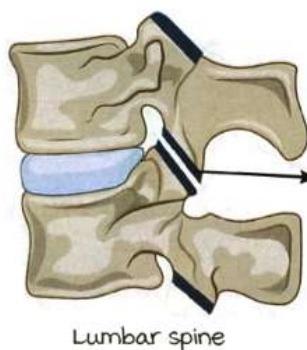
00:10:58

m/c site of

- vertebral fracture : Lower thoracic (T12)
- vertebral dislocation : Cervical spine d/t

Stabilizers : Ligamentous > bony.

Horizontal arrangement of articular facets.



Lumbar spine

Vertical arrangement of articular facets
 ↓
 ↑ risk of fracture > dislocation



Cervical spine

Highest risk of spinal cord injury : Cervical spine fracture.

Note :

Immediate management in cervical spine injury :

1. Turn patient onto their back.
2. Support placed behind neck.
3. Transported to nearest hospital.

JEFFERSON'S/C1 VERTEBRAL (ATLAS) FRACTURE

- m/c fracture on C1 vertebra.
- mechanism of injury : Axial loading/compression

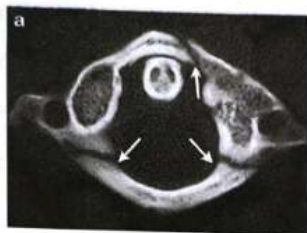
Fragments burst out away from spinal canal.

- No neurological deficits.

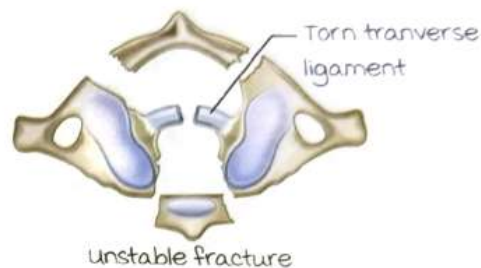
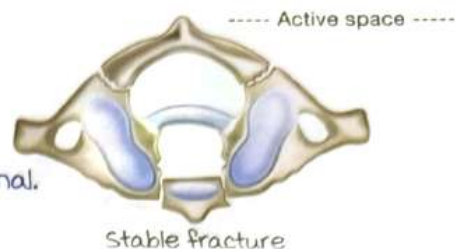
Investigations :

x-ray (open mouth view)

CT scan (best)



Treatment → Stable fracture : Conservative.
→ unstable : Surgery (Atlanto-occipital fusion)



C2 VERTEBRAE (AXIS) FRACTURE

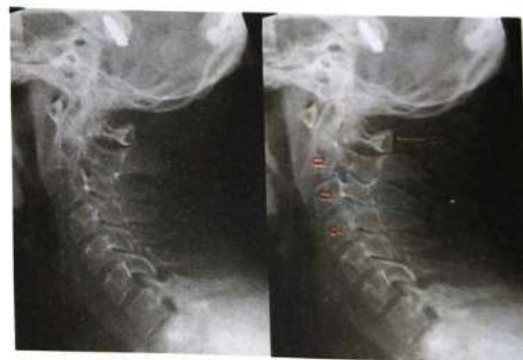
Types → Odontoid fracture : m/c fracture of C2.
→ Hangman's fracture.

Hangman's Fracture :

- Fracture of pars inter-articularis of C2 vertebra (Spondylolysis of C2)

causes dislocation of C2 over C3 (spondylolisthesis)

- No neurological deficit.
- m/c cause : RTA > Hanging.



Hangman's fracture-dislocation

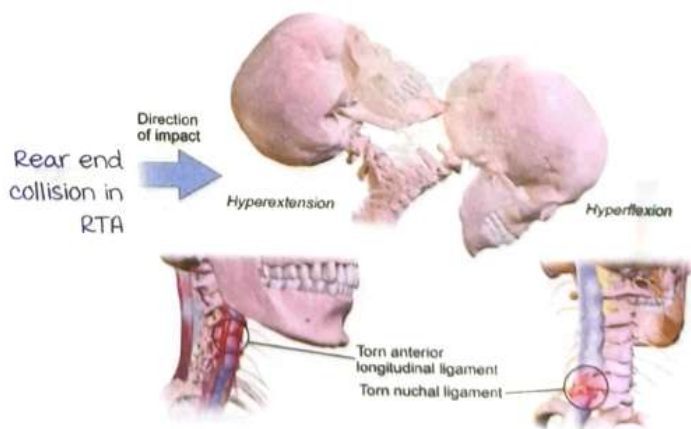
WHIPLASH INJURY

Soft tissue injury.

mechanism : Hyperextension → Flexion.

C/f :

- Chronic neck pain.
 - No neurological deficits.
 - Normal x-ray/CT scan.
- Rx : Cervical collar.



----- Active space -----

CLAY SHOVELER'S FRACTURE

Shoveling
 ↓
 Strong contraction of paraspinal
 and upper limb muscles
 ↓
 Avulsion of C7 spinous process



Clay shoveler's fracture

CHANCE/JACK-KNIFE/SEAT-BELT FRACTURE

- Described by Dr. G.Q.Chance.
- Complete transverse splitting of
 - vertebra : Bony Chance
 - Ligament & disc : Soft tissue Chance

mechanism of injury :

Seatbelts without shoulder harnesses

Flexion of spine at lumbar vertebra

Complete transection of spinal cord
 (involves all 3 columns : unstable fracture)

Paraplegia (paralysis of lower limbs)



mechanism of jackknife fracture Chance fracture

Rx : Stabilisation & surgery

COMPRESSION VS BURST FRACTURES

	Compression Fracture	Burst Fracture
Causative factor	Osteoporosis	Significant trauma
mechanism	Flexion → compression	Axial loading
Characteristic features	Wedging of vertebra (anterior column shortened, posterior column maintained) 	Convex anterior & posterior walls 

SCIWORA

00 :26 :00

----- Active space -----

- Spinal cord injury without radiographic abnormality.
- In **children** : Bone elasticity > spinal cord elasticity → Stretching of bone > spinal cord → Injury of spinal cord without fracture → Paraplegia/quadruplegia.
- Diagnosis : MRI.
- No fracture visible on X-ray/CT scan.

Spinal Cord Injury

00 :27 :31

EARLY MANIFESTATIONS

Spinal shock :

Temporary complete cessation of functions of spinal cord **below the level of injury**.

Anesthesia

Flaccid paralysis

Loss of bowel & bladder control

Loss of reflex activity

Recovery time : 24-48 hours (to weeks).

Bulbocavernosus Reflex :

- **First reflex to recover** (marks end of spinal shock).
- Root value : S2, S3, S4

Note : If Foley's catheter in situ → tug on catheter → anal sphincter contracts.

Neurogenic shock :

manifestation of neurological deficits in spinal shock.

Pathophysiology :

Loss of sympathetic outflow → hyperactive parasympathetic outflow

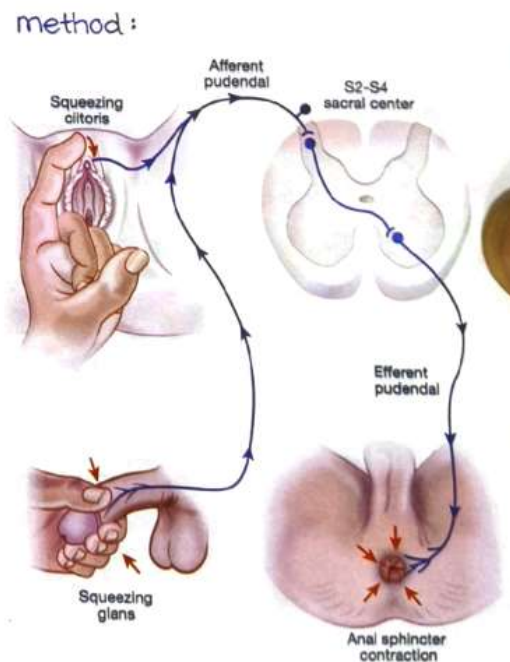
↓
Bradycardia

↓
Peripheral vasodilation

↓
↓ peripheral vascular resistance

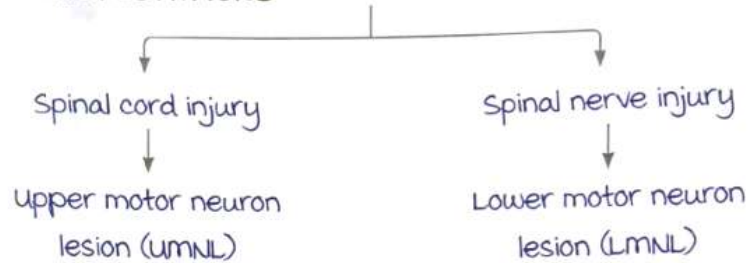
↓
Peripheral pooling of blood

↓
Fall in BP (hypotension)

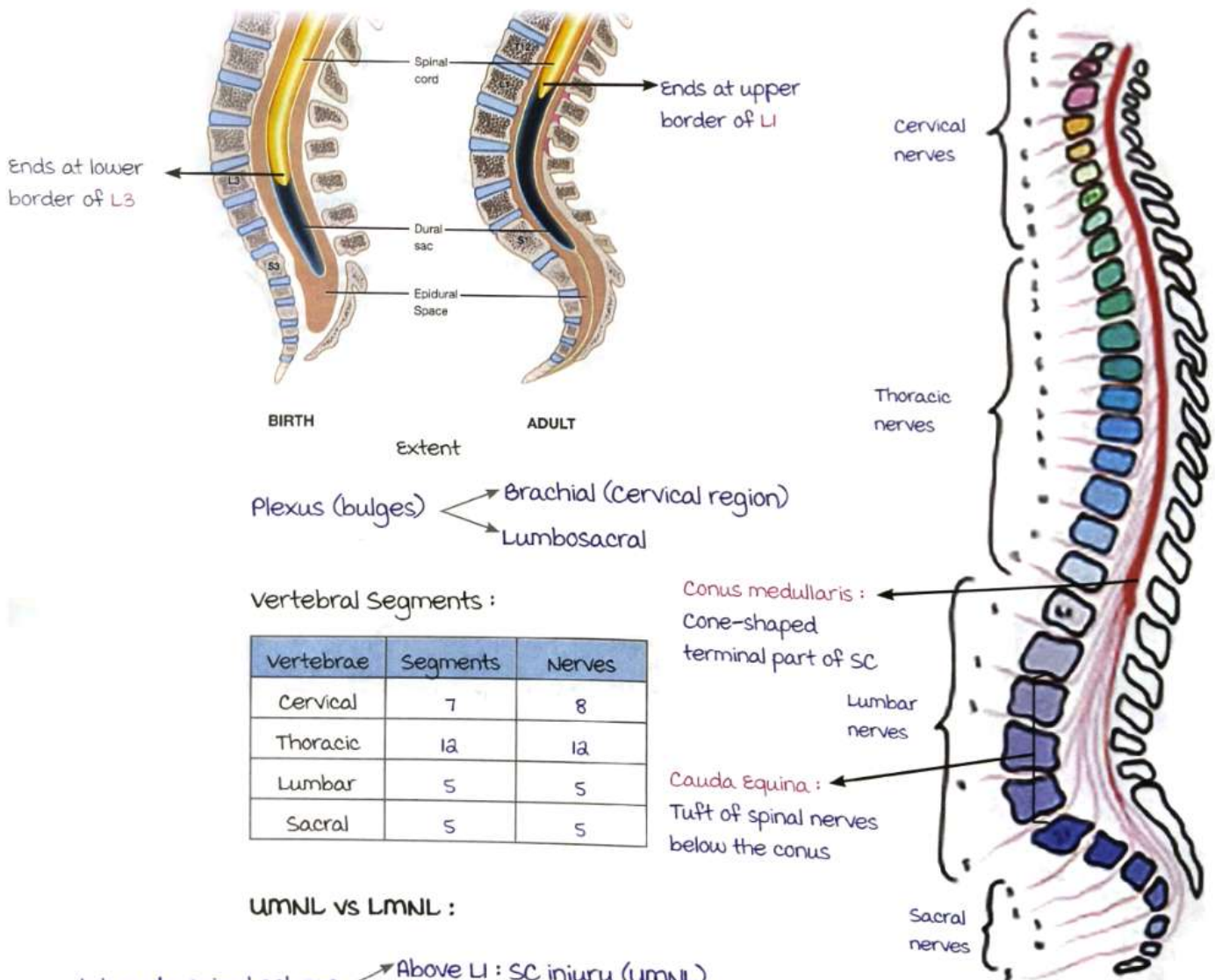


----- Active space -----

Note :

In hemorrhagic shock (blood loss) $\rightarrow \downarrow BP \rightarrow$ reflex tachycardia.**LATE MANIFESTATIONS****Relevant anatomy :**

Spinal cord :



Plexus (bulges) $\rightarrow$ Brachial (Cervical region)
 $\rightarrow$ Lumbosacral

Vertebral Segments :

Vertebrae	Segments	Nerves
Cervical	7	8
Thoracic	12	12
Lumbar	5	5
Sacral	5	5

UMNL vs LMNL :

Injury to spinal column $\rightarrow$ Above L1 : SC injury (UMNL)
 $\rightarrow$ Below L1 : Spinal nerve injury (LMNL)

Spinal nerve roots exit under their corresponding spinal segment pedicle.

----- Active space -----

		UMNL (Spinal cord)	LMNL (Spinal nerve)
At the level of injury :	• Sensory functions	Lost	
	• Motor functions	motor weakness (flaccid paralysis)	
	• Reflexes	Absent	
Below the level of injury :	• Sensory functions	Lost	Normal
	• Motor functions	Spastic paralysis	Normal
	• Reflexes	Exaggerated	Normal

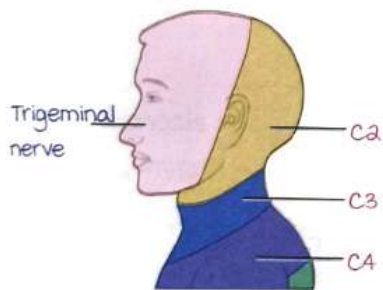
- Above the level of injury : All functions normal in both UMNL & LMNL.
- UMNL : Inhibitory action on LMN → In UMNL : Spastic paralysis with exaggerated reflexes.

Localization of injury :

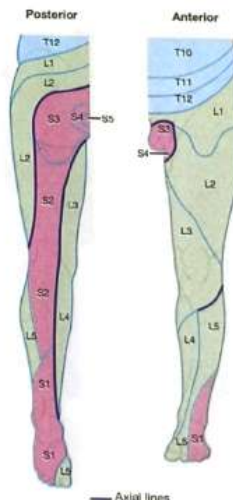
Each spinal nerve supplies a particular

- Dermatome (cutaneous distribution)
- myotome (muscle groups)
- Reflex

Dermatomes : Cutaneous sensory supply of spinal nerves.



Sensory supply of head and neck



— Axial lines

myotomes :

Root	myotome
C5	Elbow flexion (Biceps, Brachialis)
C6	wrist extension
C7	Elbow extension & wrist flexion
C8	Finger flexion
T1	Finger abduction
L2	Hip flexion/Iliopsoas
L3	Knee extension/Quadriceps
L4	Ankle dorsiflexors/Tibialis anterior
L5	Toe extensors/Extensor hallucis longus
S1	Ankle plantar flexors/Gastro soleus

----- Active space ----- Reflexes :

Root	Reflex
C5	Biceps
C6	Supinator (Brachioradialis)
C7	Triceps
L3 L4	Knee (Quadriceps)
S1	Ankle (Gastro soleus)

Note :

- Diaphragm : Nerve supply C3, C4, C5.
- Injury above C3 → affects breathing.

SPINE : DISORDERS

----- Active space -----

Prolapsed Intervertebral Disc (PIVD)

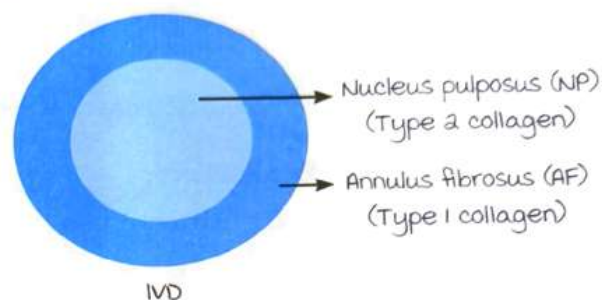
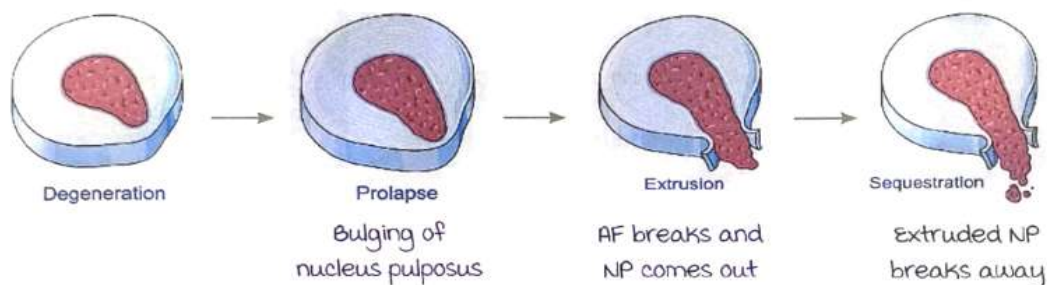
00:00:11

- Protrusion of disc $\rightarrow$ Compression of spinal cord/nerves.
- m/c disc to prolapse : $L_4-L_5 > L_5-S_1$.

mechanism of injury :

- Abnormal loading of spine.
- Degeneration of spine.

Progression of disc degeneration :

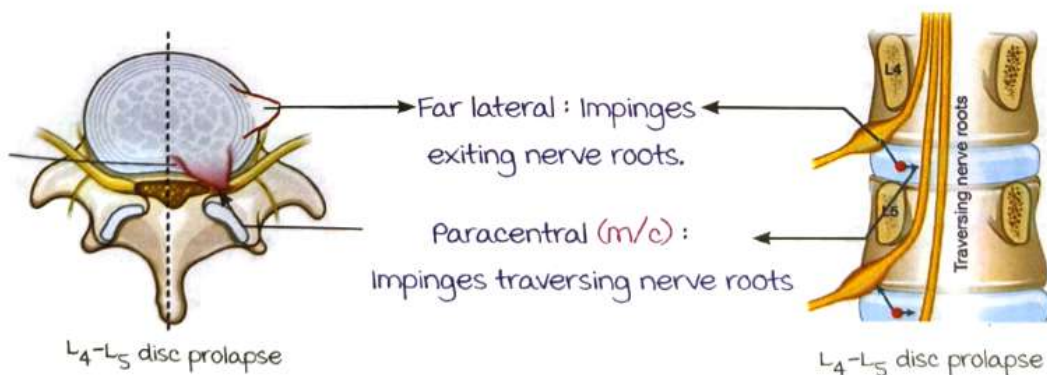


Canal stenosis :

Causes $\rightarrow$ PIVD
 $\rightarrow$ vertebral degeneration in OA $\rightarrow$ Excessive growth of abnormal bone
 (m/c site : Lumbar)

	PIVD	Canal stenosis
Age at presentation	Young	Elderly
Onset of disease	Acute	Chronic

Types of PIVD :



m/c injured nerve : L_5 (LMNL d/t nerve root involvement).

----- Active space -----

Clinical features :

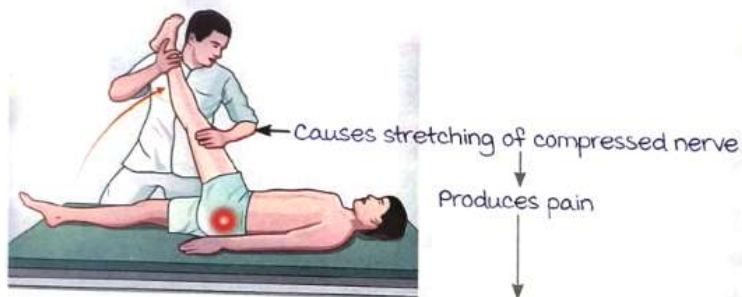
- H/o lifting heavy objects.
- Low back ache.
- Sciatica
 - Back pain radiating to lower limb.
 - Increases with
 - Flexion
 - Sneezing
 - Coughing
 - Sitting for prolonged duration
- Neurological deficits.

Note :

m/c cause of sciatica : PIVD.

Evaluation :

SLRT (Straight Leg Raising Test)

Unable to flex hip $>20-30^\circ$ (Positive test)

MRI (IOC) :

L₅-S₁ PIVD

Neurological examination : To localize the lesion depending on nerve roots involved.

Root	myotome
C ₅	Elbow flexion
C ₆	Wrist extension
C ₇	Elbow extension & wrist flexion
C ₈	Finger flexion
T ₁	Finger abduction
L ₂	Hip flexion/Iliopsoas
L ₃	Knee extension/Quadriceps
L ₄	Ankle dorsiflexors/Tibialis anterior
L ₅	Toe extensors/Extensor hallucis longus
S ₁	Ankle plantar flexors/Gastrosoleus

Root	Reflex
C ₅	Biceps
C ₆	Supinator (Brachioradialis)
C ₇	Triceps
L ₃ , L ₄	Knee (Quadriceps)
S ₁	Ankle (Gastrosoleus)

Treatment :

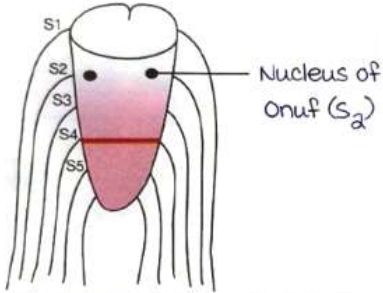
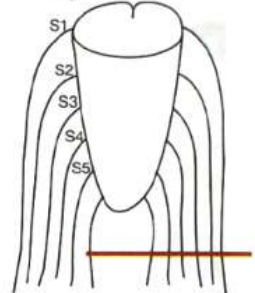
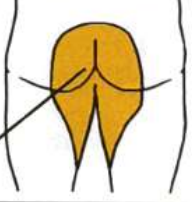
1. Rest in semi-fowler position (Supine with $30-40^\circ$ flexion of spine).
2. Physiotherapy.

3. Surgical disc compression :

- Total/hemi-laminectomy or laminotomy & discectomy.
- Indications :
 - Worsening symptoms despite conservative Rx.
 - Bowel & bladder involvement.
 - massive central disc prolapse compressing multiple nerve roots.
 - Cauda equina syndrome.

----- Active space -----

Conus medullaris v/s cauda equina syndrome :

	Conus medullaris	Cauda equina syndrome
Structure involved	 Spinal cord Nucleus of Onuf (S₂)	 Spinal nerves
Presentation	UMNL with LMNL features	LMNL
	 • B/I symptoms • Bowel & bladder involvement • Saddle anaesthesia	
	Symmetric, areflexic flaccid paralysis	Assymetrical, areflexic flaccid paralysis
Etiology	Injury at L ₁	Large disc prolapse below L ₁
Knee reflex	Normal	Lost
Ankle reflex	Lost	Lost

Lumbar Canal Stenosis (LCS)

00:24:06

Pathophysiology :

Spondylosis : Osteoarthritis of facet joints

↓

Excessive bone formation

↓

Stenosis of lumbar canal

↓

Neurogenic claudication



Lumbar canal stenosis

----- Active space -----

Clinical features :

Pain → Decreases with flexion : Sitting, leaning over, walking upstairs/uphill.
 → Increases with extension : Walking, standing upright

Neurogenic v/s vascular claudication :

Symptoms	Neurogenic	Vascular
walking (upright)	Causes symptoms	
Sitting	Relieves symptoms	
Pain	Starts in back	Starts in calf
Standing stationary	Causes symptoms	Relieves symptoms
Climbing upstairs	Relieves symptoms	Causes symptoms
Postural changes	Yes	No
Pulses	Normal	Abnormal

LCS v/s PVD :

	LCS	PVD
Age at presentation	Elderly	Young
Onset of disease	Slow & progressive	Acute
Pain on flexion	Decreases	Increases

Spondylolisthesis

00:28:26

- Slipping of 1 vertebrae over another.
- Cause : Trauma/fracture of the pars interarticularis.

Radiological features :

Normal spine
(Scottie dog sign)



Spondylolisthesis
(Beheaded scottish
terrier sign)
m/c site : L₅-S₁



Spondylolysis
(Scottish terrier with
collar sign)
m/c site : L₅



Spondylolysis

Meyerding classification:
Based on percentage of slip.

----- Active space -----



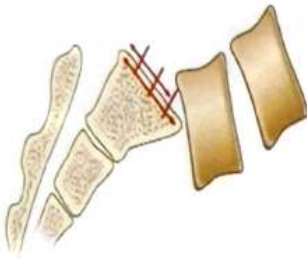
Normal



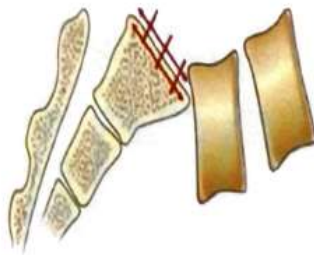
Grade I : 0-25% displacement



Grade II : 26-50% displacement



Grade III : 51-75% displacement



Grade IV : 76-100% displacement



Grade V : >100% displacement
(Spondyloptosis)

Clinical features :

- Back pain radiating to the lower limbs.
- Step sign : On palpation of spine
↓
Step off over the spinous process



Step sign

Spondylolisthesis

Note :

- Spondylosis : Degeneration of the spine (Elderly).
- Spondylitis : Inflammation seen in ankylosing spondylitis.
- **Inverted Napoleon hat sign** (In spondyloptosis).



Inverted Napoleon hat sign

----- Active space -----

Conditions With Fusion of Spine

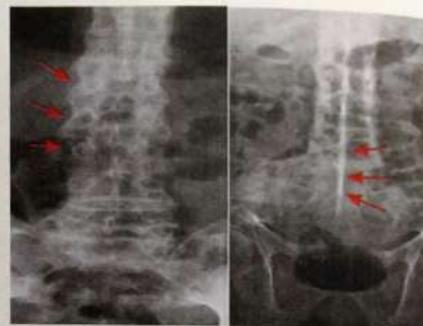
00:32:24

Ankylosing spondylitis

- Autoimmune.
- Males > females, young patient.
- HLA-B₂₇ (+) ; Rheumatoid factor (-).
- Pathology : Enthesopathy.

Clinical features :

- Characteristic feature : **Sacroiliac joint involvement** (Sacroiliitis).
- Morning stiffness (inflammatory exudates collect at night) → Improves with activity.
- Associated anterior uveitis.



Bamboo spine

Dagger sign

Radiological features (X ray)

- **Bamboo spine** : Enthesitis → vertical syndesmophyte fusion.
- Trolley track sign } D/t calcification of ligaments.
- **Dagger sign**

Diffuse idiopathic skeletal hyperostosis (DISH)

AKA Forestier disease.

- Ossification of anterior longitudinal ligament.
- Age at presentation : Elderly.



DISH

OPLL

Ossification of posterior longitudinal ligament (OPLL) :

- m/c site : Cervical spine.
- Age at presentation : Elderly.

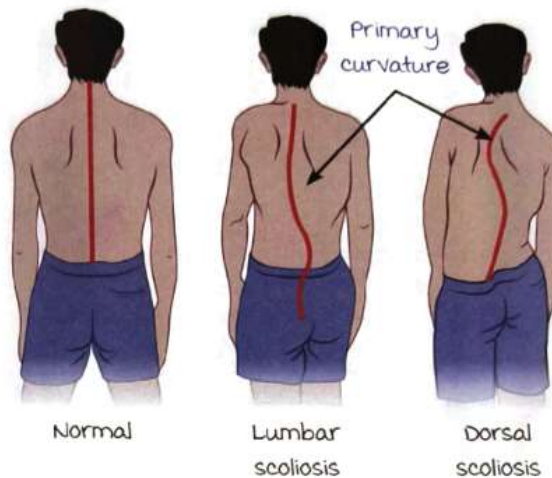
Scoliosis

00:35:17

Lateral deviation of spine by $>10^\circ$ from central axis.

SPINAL CURVATURES

- Primary curvature : The largest curve.
- Secondary curvature : Other smaller curves above & below primary curve.



----- Active space -----

Types of deviation of spinal column

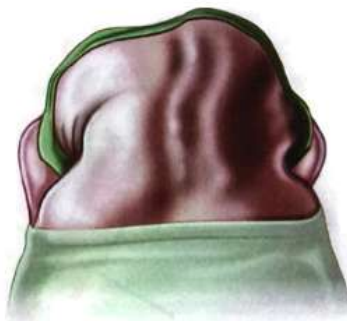
- Lateral.
- Sagittal (Forward bending).
- Rotational.

Scoliosis

- Structural (Fixed).
- Non-structural (Postural/compensatory) : No rotational component.

Adam's forward bending test :

Screening test to differentiate structural v/s non-structural.

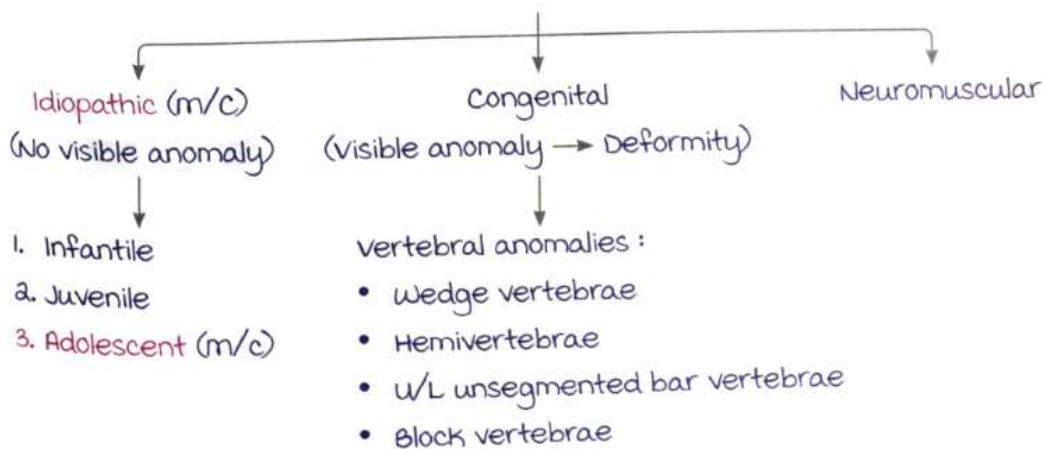


Structural scoliosis
Scoliosis persists on bending forward



Non structural scoliosis
Scoliosis disappears on bending forward

CLASSIFICATION OF STRUCTURAL SCOLIOSIS



----- Active space -----



Congenital scoliosis



Wedge vertebrae



Block vertebrae

CLINICAL FEATURES

- Asymmetry of shoulder.
- Lateral deviated spine.
- Rib hump d/t rotational abnormality.
- Deformity → Overcrowding of ribs → Compresses thoracic and abdominal structures.

EVALUATION

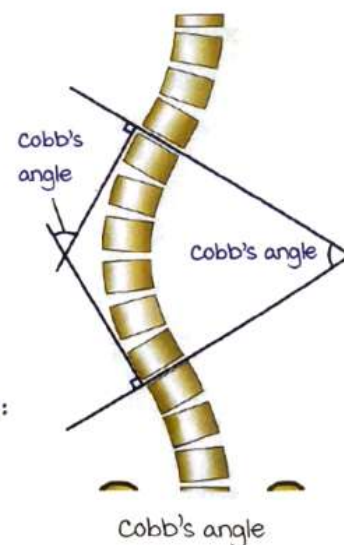
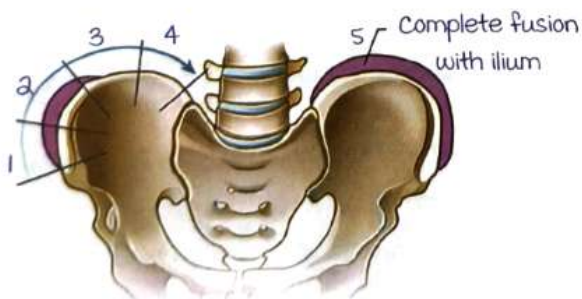
Severity of scoliosis :

Cobb's angle : Angle between abnormal vertebrae on both ends of primary curvature.

Prognosis :

Risser sign :

- To assess progression potential (α growth potential : High in children, low in adults).
- Determine skeletal maturity based on extent of fusion of iliac apophysis with iliac crest.



Risser grade	Description
Grade 1	Ossification of lateral 25%
Grade 2	Ossification of lateral 50%
Grade 3	Ossification of lateral 75%
Grade 4	Ossification of lateral 100%
Grade 5	Fusion with the ilium

TREATMENT

----- Active space -----

Depending on severity & prognosis

- Low cobbs angle (Low severity)
- High Risser (Good prognosis)

- High cobbs angle (High severity)
- Low Risser (Bad prognosis)

Conservative (Braces)

Surgical correction :



milwaukee brace
(Cervical-thoraco-lumbo-sacral orthosis)



Boston brace
(Thoraco-lumbo-sacral orthosis)



Charleston brace



Pedicle screw fixation with
Harrington rods
(Screw through pedicle involves
all 3 spinal columns)

Note :

- Incidence : Females > males.
- Prognosis : males > females.

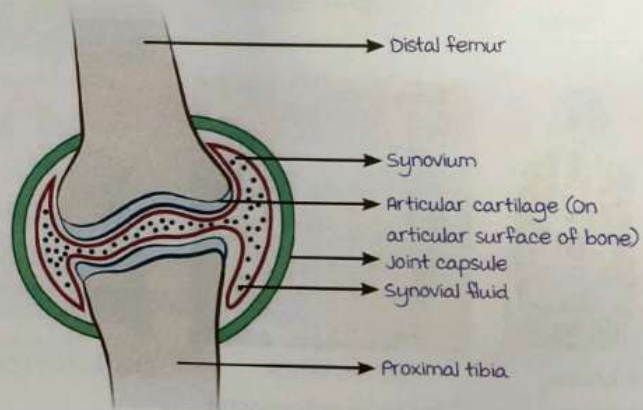
----- Active space -----

JOINT DISORDERS

Anatomy of a Joint

00:01:45

PARTS OF KNEE JOINT



Damage to articular cartilage

↓
Painful movement (active & passive) of the joint

Joints involved in osteoarthritis and rheumatoid arthritis

	Osteoarthritis	Rheumatoid arthritis
Joints involved	Distal interphalangeal joint (DIP) Proximal interphalangeal joint (PIP) 1 st carpometacarpal joint	MCP Wrist PIP
Joints spared	Wrist Metacarpophalangeal joint (MCP)	DIP



Osteoarthritis

00:06:38

m/c joint disease.

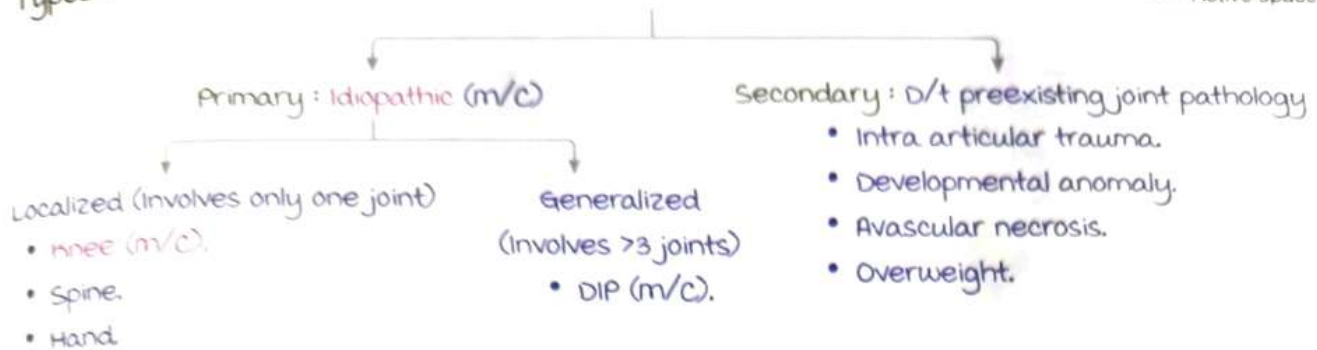
ETIOPATHOGENESIS

Etiology :

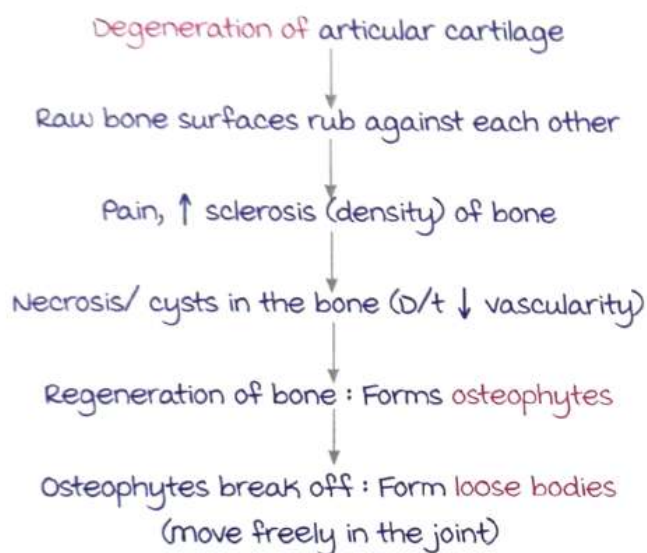
1. Age related degeneration.
2. Preexisting destruction of cartilage.

Types :

----- Active space -----



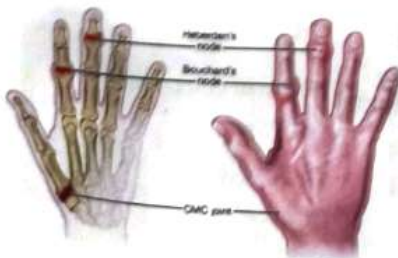
Pathogenesis :



FEATURES

Clinical features :

- Pain on active & passive movement.
- ↓ range of movement → ↓ muscle usage → Disuse atrophy (m/c : vastus medialis of quadriceps)
- Crepitations.
- Swelling and deformity.
 - Bouchard's and Heberden's nodes : D/t osteoarthritis of PIP and DIP.



DIP swelling (Heberden's node)
PIP swelling (Bouchard's node)



----- Active space -----

X-ray features :

1. Decreased joint space.
2. Bone on bone appearance.
3. Subchondral sclerosis.
4. Necrotic cysts & osteophytes.
5. Loose bodies.



Osteoarthritis



Normal Knees

Note :

- most of the weight bearing occurs on the medial side → Joint space decreases medially first → Genu varum.
- Osteoarthritis is the m/c cause of b/l genu varum.

Loose bodies in knee joint :

- m/c cause : Osteoarthritis.
- m/c cause in elderly : Osteoarthritis.
- m/c cause in young : Osteochondritis dissecans.
- m/c cause of multiple loose bodies : Synovial chondromatosis.

TREATMENT**1. Conservative :**

- Prevent progression : Weight loss.
- Physiotherapy.
 - Heat.
 - Strengthening exercises.
- Walking with support.
- Braces.

2. Medication (Supportive treatment) :

- NSAID's : Acetaminophen.
- Cartilage protectors : Glucosamine, chondroitin sulphate
- Help regenerate damaged cartilage in early stages.
- Lubrication : Inj. hyaluronic acid (Visco supplements).

3. Surgical (Definitive) :

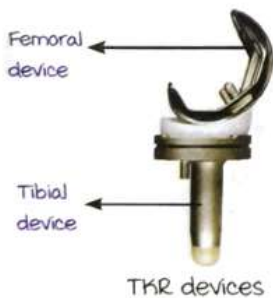
Arthroscopy : Joint wash to remove loose bodies.

High tibial/ medial opening wedge osteotomy : To correct varus deformity & ↓ pain.

Arthroplasty : To replace damaged joint.

1. Unicondylar replacement : Replacement of medial side of knee joint.

2. Total knee replacement (TKR) → Complication : Patellar clunk syndrome.



TKR devices



High tibial osteotomy



Unicondylar replacement



- Femoral component replacement
- Tibial component replacement

3. Total hip replacement :

- Head & neck of femur replaced.
- Acetabular component.



Total hip replacement

----- Active space -----

Rheumatoid Arthritis

00:18:54

INTRODUCTION

General features :

- m/c type of inflammatory arthritis disease of synovium.
- Systemic autoimmune multisystem disease (Lungs, heart, liver).
- Predominantly musculoskeletal manifestations.
- Erosive type of arthritis → Destroys joint.
- Females > males.
- Symmetrical (B/L involvement) peripheral joint involvement.

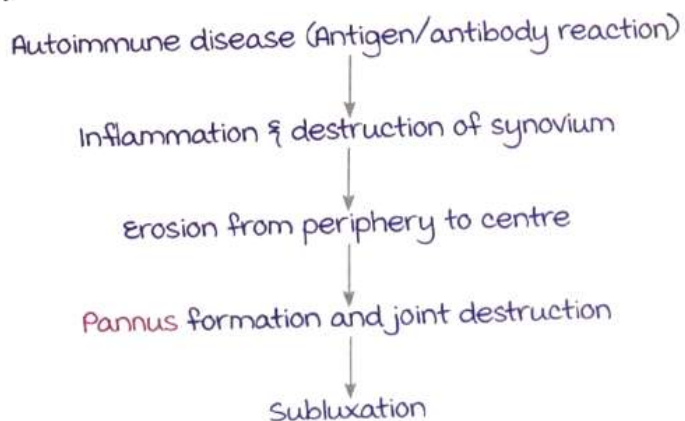
Joints involved :

- Hand : MCP > wrist, PIP.
- Knee and hip.
- Axial is rare (may involve cervical spine).
- Spares : DIP.

Note : Only non-erosive inflammatory arthritis is SLE.

PATHOLOGY

Pathogenesis :



----- Active space -----

multisystem associations :

↓ life expectancy

- Carditis.
- Pleuritis.
- Caplan syndrome.
- Felty syndrome.

INVESTIGATIONS**Diagnosis :**

- ↑ ESR, ↑ CRP.
- Rheumatoid factor : **most sensitive** marker.
 - IgM antibody against FC portion of IgG Ab.
- **Anti cyclic citrullinated peptide (Anti-CCP)** : **most specific**.

X-ray features :

1. Narrowing of joint space.
2. Osteoporosis (↓ bone density) and hyperemia :
 - D/t inflammation of joint.
 - Initially juxta articular, later generalised.
3. marginal erosion.
4. **No sclerosis, no osteophytes.**



Rheumatoid arthritis

Normal joint

Difference b/w OA and RA :

RA

OA

Note :

DeQuervain's synovitis : Tenosynovitis of abductor pollicis longus and extensor pollicis brevis.

DEFORMITIES IN RA

1. Hand :

----- Active space -----

ulnar deviation
of fingersRadial deviation
of metacarpals

1. Z deformity



2. Boutonniere deformity



3. Swan neck deformity

Boutonniere
deformitySwan-neck
deformity

Flexion of PIP & compensatory extension of DIP

Extension of PIP & compensatory flexion of DIP

Note : In mallet finger, only injured finger is affected (DIP flexed).

2. Knee :

- B/L genu valgus.
- Wind swept deformity.
- Cause : Rickets > RA in children
RA in elderly.
- Triple deformity.

3. Foot :



1. Hammer toes



2. Hallux valgus

Note : ulnar nerve palsy → Claw hand.

MANAGEMENT

1. Physiotherapy : Preserves and improves joint function.

2. methotrexate : DOC.

3. DMARD's :

- Leflunomide : Immuno suppressant.
- Infliximab/Adalimumab : TNF- α inhibitors.
- Anakinra : IL-1 receptor antagonist.
- Rituximab : Anti CD20 antibody.

4. Surgical :

- Synovectomy : Removal of inflamed synovium.
- Joint replacement surgery.
- Deformity corrections.



3. Wind swept deformity

Spondyloarthropathies

00:32:07

Inflammatory disease processes involving spine and joints.

Conditions :

1. Ankylosing spondylitis : m/c.
2. Psoriatic arthritis.
3. Enteropathic arthritis : Inflammatory bowel disease (Crohn's, ulcerative colitis).
4. Reactive arthritis : Shigella, chlamydia.

Features :

- They are seronegative (RF -ve).
 - Young males > females.
 - HLA B27 positive.
 - A/w uveitis.
- Note : RA is a/w scleritis.

ANKYLOSING SPONDYLITIS

AKA *Marie Strumpell disease* / *Bechterew's disease*.

General features :

- males > females.
 - Joint involvement : Axial > peripheral.
- Involves :
- Sacroiliac joint (m/c).
 - Spine.
 - Hip.

Clinical features :

- Pain at gluteal region/back (Sacroiliitis).
- *morning stiffness* improving with activity.
- Decreased lumbar spine movement.
- ↓ chest expansion (D/t fusion of ribs with the spine).
- Anterior uveitis.

Pathogenesis :

Enthesopathy (Enthesis : Site of insertion of ligaments & tendons on bone).

Inflammation : Enthesitis → Erosion → Abnormal regeneration & calcification →

Restriction of joint movement.

X-ray features :

1. Sacroiliitis :
Sclerosis and ankylosis (fusion) of sacroiliac joint.



Normal sacroiliac joint

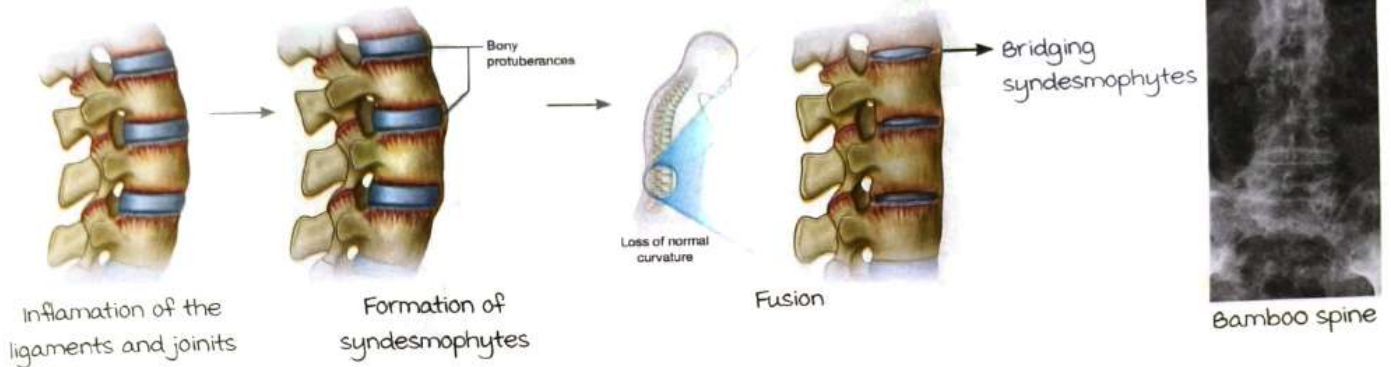


Ankylosis & sclerosis

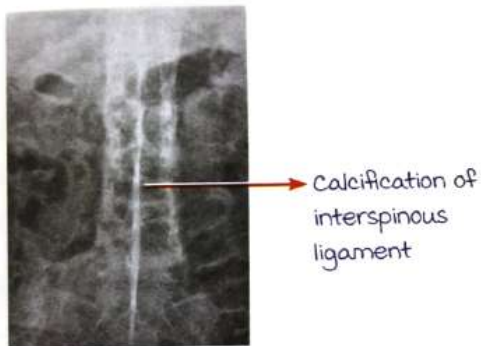
----- Active space -----

2. Bamboo spine :

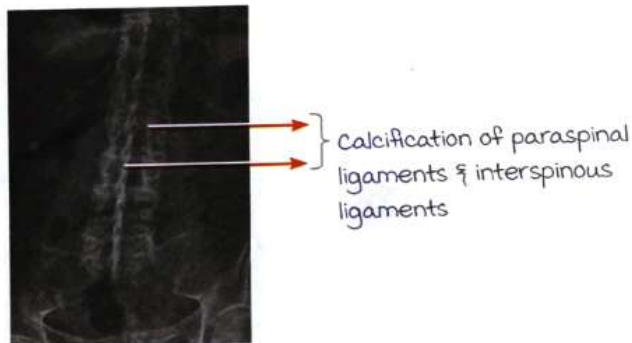
- Fusion and squaring of spine.
- Calcification b/w vertebrae.



3. Dagger sign :



4. Trolley track sign



Diagnostic criteria :

1. Essential criteria : Sacroiliitis
 - MRI (IOC) : most sensitive (Earliest changes seen).
 - Inflammation of iliac > sacral side.
2. Supportive criteria :
 - ↓ lumbar spine movement.
 - ↓ chest expansion.
 - Inflammatory back pain & morning stiffness.

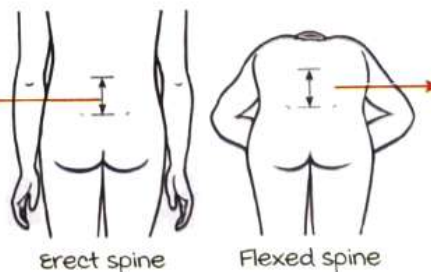
Clinical tests :

- ↓ chest movement (measure chest expansion)
- For sacroiliitis :
 - Gaenslen's test.
 - FABER test/ figure of 4 test.
 - Pump handle test.
- Testing lumbar spine motility :
Schober/ modified schober test.

Treatment :

1. Chest physiotherapy.
2. Exercises : Swimming, cycling.
3. Medical management :
 - NSAID's.
 - Steroids.

1. measure distance b/w 2 points



2. measure same distance on flexion : Spine does not open upon flexion (d/t fusion). The distance b/w 2 points does not ↑.

----- Active space ----- Note :

Differential diagnosis for fusion of spine

1. DISH : Diffuse idiopathic skeletal hyperostosis

- AKA forestier disease.
- m/c in elderly.



Anterior longitudinal ligament calcification :
Flowing candle wax like ossification

2. OPLL : Ossification of posterior longitudinal ligament

- m/c in elderly.
- m/c in cervical spine.



Ossification of posterior longitudinal ligament

PSORIATIC ARTHRITIS

General features :

- Erosive arthritis.
- Females > males.
- Asymmetrical oligoarticular type (m/c involved : DIP).

CASPAR criteria :

- Psoriasis.
 - RF -ve.
 - X-ray changes.
 - Dactylitis.
 - Nail changes.
- } For diagnosis

Clinical features :

1. Dactylitis :

Sausage digit d/t swelling of fingers.



2. Shortening of digits :

- d/t destruction of the joints.
- Telescoping (pulling out) of fingers.



3. Arthritis mutilans :

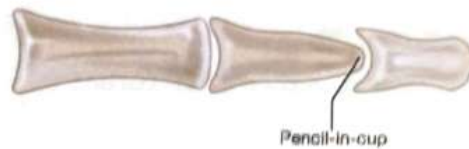
Total erosion and destruction of bone.

Treatment :

- Treatment of psoriasis.
- methotrexate.

4. Pencil in cup appearance :

Extreme destruction of phalanx.



----- Active space -----

Hemophilic Arthropathy

00:45:50

etiology :

- Hemophilia A : Factor 8 deficiency.
- Hemophilia B : Factor 9 deficiency.

Pathogenesis :

Recurrent bleeding into the joint → Destruction of synovium + inflammation.



Hemophilic arthropathy

Clinical features :

1. Pseudo tumors :

- muscular bleeding → Collection of blood in the muscle.
- m/c : Iliopsoas > quadriceps.

2. Painful swelling of joints :

- D/t bleeding into it.
- m/c : knee.
- Present in flexed position : As its position of ease.

3. Arthritis mutilans.

X-ray :



1. Widening of intracondylar notch
2. Loss of joint space

3. Juxta articular osteopenia.
4. Squaring of patella.

Treatment :

- Treat hemophilia : Factor replacement.
- Compression bandage.
- Ice pack.
- Aspiration of the joint :
 - Only in severe pain.
 - Can cause infection.

----- Active space -----

Crystal Deposition Arthropathy

00:48:00

GOUT

Features :

- m/c crystal arthropathy. uric acid levels
- Purine metabolism abnormality.
- ↑ serum uric acid (Normal : 3.5 - 6.5 mg/dl).
- m/c involved : 1st MTP (Great toe).
- male (middle age) > females.
- Drugs : Furosemide, aspirin.

Pathogenesis :

Sudden change in → Crystal precipitation and deposition in cold peripheral joints
 ↓
 Local inflammatory reaction
 ↓
 Collateral damage to the joint



Acute gout

Painful swelling : Tophi

Investigations :

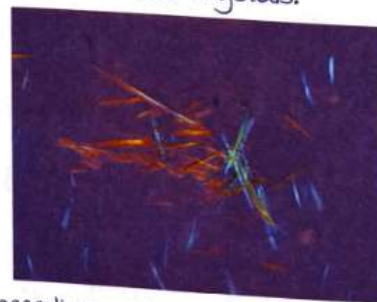
1. X-ray :

- Punched out lesion.
- Joint destruction.
- Overhanging bone/ martel & sign.



2. Synovial fluid analysis is :

- Best investigation.
- Light microscopy of joint aspirate.
- Characteristics of crystals :
 - Needle shaped, -ve birefringent, mono sodium urate crystals.



Gout : monosodium urate crystals on light microscopy

Treatment :

1. Acute gouty attack :

- Only NSAIDs : Indomethacin (Aspirin is C/D).
- Colchicine : Causes GI disturbances.
- Do not control serum uric acid during acute attack, as it can precipitate more pain.

a. Chronic gout :

a. Xanthine oxidase inhibitors :

- Allopurinol.
- Febuxostat.
- Rasburicase.

b. Uricosuric drugs :

- Probenecid.

----- Active space -----

PSEUDOGOUT

Deposition of calcium pyrophosphate dihydrate.

General features :

- m/c in elderly patients.
- A/w hypothyroidism.
- Female > male.
- m/c affects knee joint (Large joints).

X-ray features :

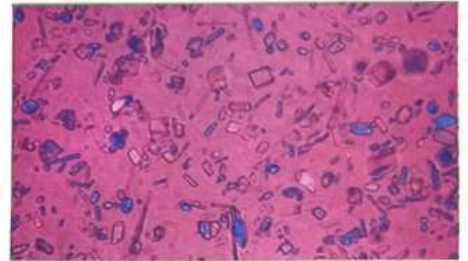
Chondrocalcinosis.



Note :

Calcification of cartilage is also seen in :

- Ochronosis / alkaptonuria.
- Hemochromatosis.
- Hyperparathyroidism.



Synovial fluid study :
Polygonal, +ve birefringence

Neuropathic Joint

00:54:10

Joint destruction d/t sensation & proprioception.

AKA Charcot joint.

Etiology :

1. Peripheral neuropathy : m/c cause.
 - Diabetes mellitus (m/c cause of peripheral neuropathy).
 - m/c involves : mid tarsal joints > Tarsometatarsal joint.
2. Leprosy.
3. Syphilis : Involves knee.

Features :

- m/c involves peripheral joints of lower limb.

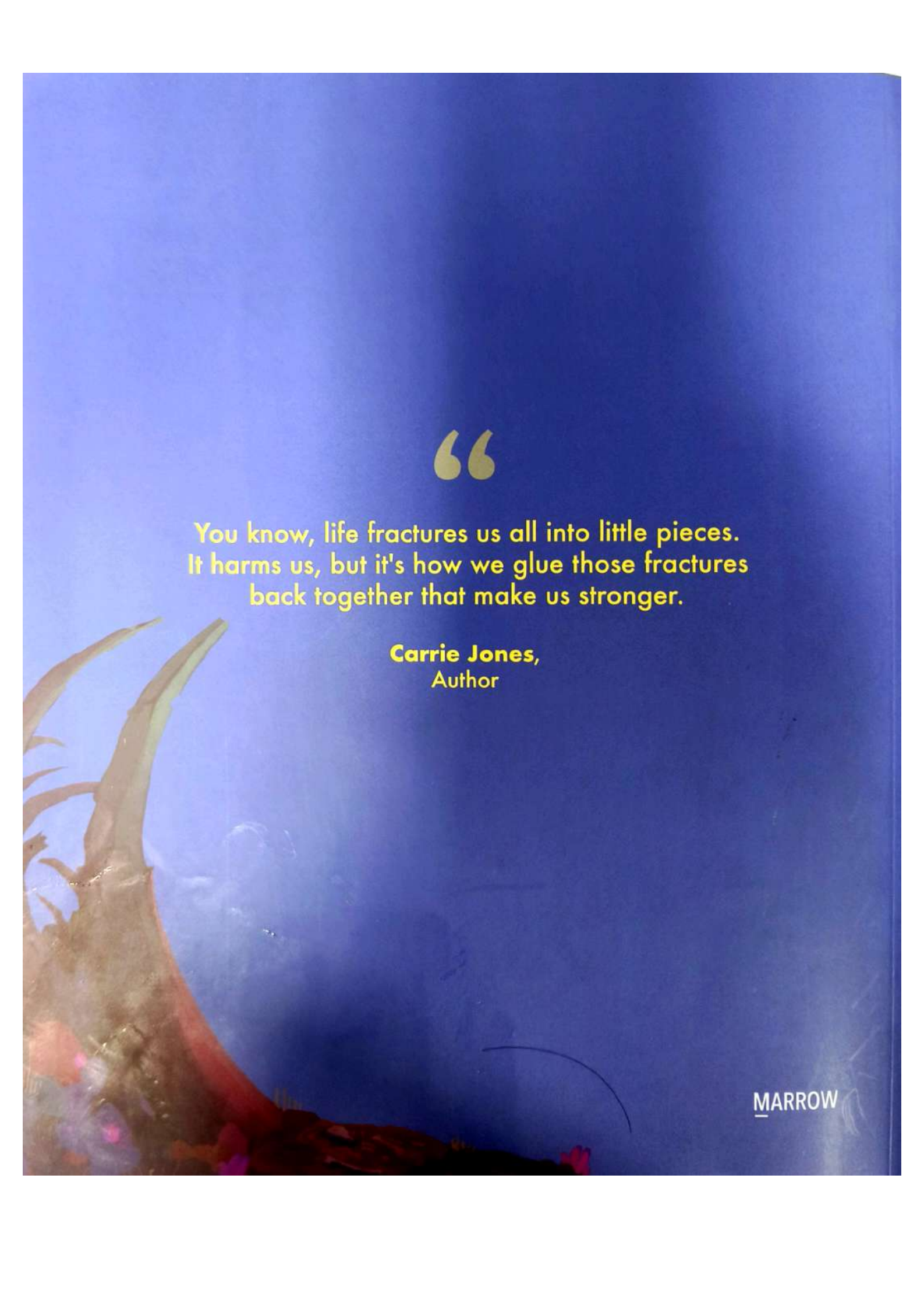
- Bag of bones appearance.
- No pain.

Treatment :

- Arthrodesis : Fusion of bones.
- Arthroplasty is C/I.



Destruction of joints & bone



“

You know, life fractures us all into little pieces.
It harms us, but it's how we glue those fractures
back together that make us stronger.

Carrie Jones,
Author

MARROW